A Project Report On

AI-Powered Room Design System

Designed By:

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Submitted To:

K.M.S.P. MANDAL'S

SANT RAWOOL MAHARAJ MAHAVIDYALAYA, KUDAL.

(NAAC Re-Accreditation "B+" Grade)

In the partial fulfillment of

B.Sc. Computer Science (Sem V)

Under Guidance Of:

Asst. Prof. Santosh Chougule (M.C.A.) | Asst. Prof. Prakash Talankar (M.C.A.)

Through:

THE HEAD, DEPARTMENT OF COMPUTER SCIENCE,

S.R.M. COLLEGE, KUDAL.

PROJECT REPORT

2026-27

DECLARATION

To,
The Head,
Department of Computer Science,
S. R. M. College, Kudal.

Respected Sir,

I, the undersigned, hereby declare that the project on “AI-Powered Room Design System (A Rule-Based Intelligent Web System for Interior Planning & 2D Blueprint Generation)” is developed under the guidance of our lecturers Asst. Prof. Santosh Chougule and Asst. Prof. Prakash Talankar.

The conclusion in this report is based on the data and source code developed by me. I declare that this is my original work and I have not copied any materials related to my project.

I do undersign that if my work is found to be copied, then I am liable to punishment as per the university rule.

DATE: _____
PLACE: KUDAL.

(Mr. Ketan Ravindra Gosavi)

K.M.S.P. MANDAL'S

SANT RAWOOL MAHARAJ MAHAVIDYALAYA, KUDAL.

NAAC RE-ACCREDITATION "B+" GRADE

CERTIFICATE

This is to certify that Mr. Ketan Ravindra Gosavi, having Exam Seat Number _____, has satisfactorily carried out his project work entitled:

AI-Powered Room Design System (A Rule-Based Intelligent Web System for Interior Planning & 2D Blueprint Generation).

As per the syllabus prescribed for B.Sc. Computer Science of Mumbai University, Mumbai.

It is also certified that this is his own work completed during the academic year 2026-27. The work done is satisfactory and is presented as per the specifications.

_____	_____	_____
Project Guide	**External Examiner**	**Head of Department**

DATE: _____

PLACE: KUDAL.

CHAPTER I

Acknowledgement

CHAPTER I

ACKNOWLEDGEMENT

It is my great pleasure to get this opportunity and sincerely thank all the people who helped me create a successful project and guided me throughout the completion of this project.

I would like to thank my project guides, Asst. Prof. S. R. Chougule (M.C.A.) and Asst. Prof. Prakash Talankar (M.C.A.), who provided important technical information and guided me throughout the development of the AI-Powered Room Design System.

I am also grateful to the Head of the Department of Computer Science, S. R. M. College, Kudal, for providing the necessary computer laboratory facilities. And last but not the least, I would like to thank all my classmates for their support.

Thank You.

Mr. Ketan Ravindra Gosavi.

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CHAPTER II

Synopsis of the Project

CHAPTER II

AI ROOM DESIGNER — AI-POWERED ROOM DESIGN SYSTEM

1. Problem Statement

Planning and designing interior rooms efficiently has become increasingly challenging for homeowners, students, and interior design enthusiasts. Although professional architectural and CAD tools exist, users often need to switch between separate applications to calculate room dimensions, choose interior styles, filter furniture within available walking space, coordinate wall and floor colors, draw 2D floor blueprints, estimate lighting lumens, and calculate renovation budgets. This fragmented approach makes interior planning time-consuming and reduces the convenience of making data-driven design decisions.

Additionally, many users struggle to determine how much floor area should be reserved for movement clearance, which color palettes match their chosen interior style (such as Modern, Luxury, Minimal, Japanese, or Scandinavian), and how many lumens are required for balanced ambient, task, and accent lighting. Manual tracking through paper sketches and calculators often leads to inconsistent measurements and budget overruns.

To address these challenges, AI ROOM DESIGNER (Blueprint Edition) — AI-Powered Room Design System is developed as a web-based application powered by HTML5, CSS3, JavaScript, and a Node.js + Express + Multer backend. It combines room photo upload, spatial dimension calculation, 5-style AI design analysis, furniture footprint planning, 4-channel color palettes, interactive 2D SVG blueprint generation, lighting simulation, and multi-tier INR budget estimation within a unified Blueprint interface.

2. Objectives and Scope

- **Centralized Room Design Workspace:** To develop a web-based application (index.html + 7 specialized stage modules) that centralizes interior room analysis, layout planning, and cost estimation.
- **Room Upload & Spatial Calculation API:** To provide a Node.js + Express + Multer backend (server.js) that validates and stores uploaded room photographs (/api/upload) and calculates room floor area and volume (/api/room/calculate).
- **Intelligent Rule-Based Style Profiling:** To support five curated interior styles—Modern, Luxury, Minimal, Japanese, and Scandinavian—and generate style-specific recommendations for furniture, colors, and lighting.
- **Ergonomic Furniture & 4-Channel Color Planning:** To reserve 35% of floor area for movement pathways (65% usable space in furniture.js) and provide six 4-channel color palettes (wall, floor, furniture, and accent hex codes in colors.js).

- **Dynamic 2D SVG Blueprint & Lighting Simulator:** To generate scaled 2D technical floor plans using SVG (blueprint.js) and simulate Ambient, Task, and Accent lighting layers with lumen and Kelvin temperature controls (lighting.js).
- **Multi-Tier INR Budget Estimator & Report Export:** To calculate itemized renovation budgets across Basic, Standard, and Premium tiers with a 5% contingency reserve (budget.js) and export a downloadable summary report (download.js).

Project Scope: Scope: The project focuses on residential room analysis, 2D SVG floor blueprint generation, lighting/color coordination, and approximate Indian Rupee (INR) renovation budgeting. Budget rates are planning estimates rather than binding contractor quotations, and AI style simulations provide structured design guidance to assist human decision-making.

3. Process Description

The existing room planning workflow requires users to manually calculate square footage, guess furniture proportions, search separately for color combinations, and estimate painting and labor costs in spreadsheets. Without a unified state management workflow, changes made to room dimensions do not automatically update blueprint scales, lumen targets, or budget totals.

AI Room Designer (Blueprint Edition) solves this through a 7-stage connected pipeline backed by Node.js/Express and browser localStorage persistence (aiRoomData, aiColorData, aiLightingData, and aiBudgetData). The user starts on index.html by uploading a room image (validated for JPG/PNG/WEBP and stored in /uploads), entering Length, Breadth, and Height in feet, and selecting one of five interior styles. Clicking 'Calculate Room' computes floor area (sq.ft), volume (cu.ft), and room classification (Small Room, Medium Room, Large Room, or Spacious Room).

From the main workspace, the user progresses seamlessly through seven specialized modules: (1) Room Analysis (analysis.html), (2) Furniture Selection (furniture.html), (3) Color Palettes (colors.html), (4) 2D SVG Blueprint (blueprint.html), (5) Lighting Setup (lighting.html), (6) Renovation Budget (budget.html), and (7) Final Summary & Download (download.html), where the complete design specification can be printed or downloaded as ai-room-design-summary.txt.

CHAPTER III

Main Report

CHAPTER III

MAIN REPORT: AI-POWERED ROOM DESIGN SYSTEM

■ Objective and Scope of the Project

AI Room Designer — Blueprint Edition is engineered to simplify interior space planning through a structured seven-stage workflow. The key objectives implemented in the system are:

Objective Area	Implemented Functionality in AI-Room-Designer
1. Unified Blueprint Interface	Provide a cohesive dark-blueprint engineering aesthetic (--blueprint-dark: #071a2e, --cyan: #29d8ff) across the home studio and all sub-pages.
2. Express & Multer Backend API	Implement server.js on Port 3000 with /api/status, /api/upload (Multer disk storage up to 5 MB), and /api/room/calculate endpoints.
3. Automated Spatial Classification	Calculate floor area (L × B) and volume (L × B × H) and classify rooms into Small (<80 sq.ft), Medium (<150 sq.ft), Large (<250 sq.ft), or Spacious (>=250 sq.ft).
4. Ergonomic Furniture Allocation	Automatically reserve 35% of room area for movement pathways and expose 65% as usable furniture footprint in furniture.js.
5. 4-Channel Color & SVG Blueprints	Synchronize 6 curated palettes (Modern Tech, Minimal Neutral, Japanese Calm, Scandinavian, Luxury Warm, Cool Blue) with a scaled 1000×650 SVG floor plan.
6. Layered Lighting & Lumen Math	Model Ambient, Task, and Accent lighting toggles with intensity scaling (baseLumens = area × 15 × intensity/70) and Kelvin color temperature.
7. Multi-Tier INR Budget & Export	Compute itemized INR budgets across Basic, Standard, and Premium tiers (Furniture, Lighting, Decor, Painting, Labour + 5% Contingency) and export .txt reports.

■ Theoretical Background

- **Client-Server Web Architecture (Node.js & Express):** The application uses Express 5 (server.js) to serve static HTML/CSS/JS assets and expose RESTful JSON endpoints. Multipart form uploads are processed by Multer middleware, which validates MIME types (image/jpeg, image/png, image/webp) and writes timestamped files to the /uploads directory.

- **Cross-Page State Persistence (Web Storage API):** To maintain a smooth multi-page workflow without requiring complex database setup for every intermediate click, the system serializes structured JSON objects into browser localStorage under dedicated keys: aiRoomData, uploadedRoomImage, aiColorData, aiLightingData, and aiBudgetData.
- **Scalable Vector Graphics (SVG) 2D CAD Rendering:** In blueprint.js, room dimensions (Length × Breadth) are dynamically scaled to fit a 1000×650 SVG viewport (scale = Math.min(680/length, 420/breadth)). Programmatic DOM creation (createElementNS) draws the background coordinate grid, outer walls, door swing arc, window opening, scaled furniture blocks, perimeter labels, and interactive zoom/grid toggles.
- **Interior Ergonomics & Lumen Illumination Formulas:** Space planning reserves 35% of total square footage for unobstructed human circulation (available = area * 0.65). Lighting estimation applies a baseline of 15 lumens per sq.ft. adjusted proportionally by the user's selected intensity slider.
- **Parametric Cost Estimation:** Renovation budgeting multiplies room square footage by tier-specific per-sq.ft. rates (Basic: INR 950/sq.ft base; Standard: INR 1,500/sq.ft base; Premium: INR 2,300/sq.ft base) across five cost heads and adds a 5% contingency reserve.

■ Survey of Literature

System / Approach	Strengths	Limitations / Gap Addressed
Desktop CAD Software (AutoCAD, SketchUp)	High drafting precision for architects.	Requires complex training; does not auto-calculate INR renovation budgets or style recommendations from simple L×B×H inputs.
Static Interior Blogs & Pinterest Boards	Visual inspiration for room styles.	No spatial calculation, no 2D blueprint scaling for user dimensions, and no cost breakdown.
AI Room Designer — Blueprint Edition (Proposed)	Integrates image upload, L×B×H calculation, 5 AI style profiles, 65% usable space rule, SVG blueprint, lighting simulator, and INR budget calculator.	Uses rule-based planning rates and simulated AI style profiles tailored for fast educational and homeowner planning.

■ Research Analysis, Design & Project Directory Structure

The project is organized into a clean modular folder hierarchy separating the root studio and Node.js server from stage-specific HTML pages, stylesheets, JavaScript controllers, style reference images, and uploaded room photos:

Project Directory Structure (AI-Room-Designer/)

```
AI-Room-Designer/
│
├── index.html                  # Main Home Studio, Image Upload, Dimensions & 5
Style Selector
├── style.css                   # Core Blueprint Dark Theme & Responsive Layout
Stylesheet
├── script.js                   # Main Frontend Logic, AI Simulation & Express API
Integration
├── server.js                   # Node.js + Express + Multer Backend Server (Port
3000)
├── package.json                # Project Metadata & Express (^5.1.0) Dependency
Manifest
│
├── images/                     # Reference Style Thumbnails
│   ├── modern.jpg
│   ├── luxury.jpg
│   ├── minimal.jpg
│   ├── japanese.jpg
│   └── scandinavian.jpg
│
├── pages/                      # 7-Stage Pipeline Pages + Login & Contact
│   ├── analysis.html           # Stage 1: AI Room Analysis & Style
Recommendations
│   ├── furniture.html          # Stage 2: Ergonomic Furniture Catalog & Cost
Tracker
│   ├── colors.html             # Stage 3: 4-Channel Interior Color Palette
Selector
│   ├── blueprint.html          # Stage 4: Interactive 2D SVG Technical Floor
Blueprint
│   ├── lighting.html           # Stage 5: Ambient/Task/Accent Lighting & Lumen
Simulator
│   ├── budget.html             # Stage 6: Basic / Standard / Premium INR Budget
Calculator
│   ├── download.html           # Stage 7: Final Design Summary, Print & .txt
Export
│   ├── login.html              # User Workspace Sign-In Page
│   └── contact.html            # Project Feedback & Contact Page
│
├── css/                        # Modular Page-Specific Stylesheets
│   ├── analysis.css, furniture.css, colors.css, blueprint.css
│   ├── lighting.css, budget.css, download.css, login.css, contact.css
│
├── js/                         # Modular Page-Specific JavaScript Controllers
│   ├── analysis.js, furniture.js, colors.js, blueprint.js
│   ├── lighting.js, budget.js, download.js, login.js, contact.js
│
└── uploads/                    # Multer Destination Directory for Uploaded Room
Photos
```

Functional Requirements Implemented

- **1. Room Image Upload & Multer Validation:** Users can drag-and-drop or browse a room image on index.html. Both client-side JS and POST /api/upload validate that the file is JPG, PNG, or WEBP and within the size limit.
- **2. Room Dimension & Volume Calculation:** Users enter Length, Breadth, and Height (in feet). Both script.js and POST /api/room/calculate verify positive numbers and compute Area = L × B and Volume = L × B × H.
- **3. 5-Style AI Design Simulation:** Users select Modern, Luxury, Minimal, Japanese, or Scandinavian style and trigger a 5-stage progress simulation (Analyzing image -> Processing dimensions -> Selecting furniture -> Generating colors -> Preparing blueprint).
- **4. Multi-Stage Navigation & State Sync:** Each stage page (analysis, furniture, colors, blueprint, lighting, budget, download) automatically reads and updates shared state in localStorage.
- **5. Interactive 2D SVG Blueprint Controls:** Users can toggle the coordinate grid, toggle furniture visibility, zoom in/out, reset view, and export the generated SVG floor plan.
- **6. Final Report Download & Print:** On download.html, users can download a formatted plain-text specification file (ai-room-design-summary.txt), trigger browser printing (window.print()), or reset localStorage to start a new design.

■ System Planning & Gantt Chart

Development of AI Room Designer — Blueprint Edition was planned across 16 weeks, progressing from UI wireframing and Express/Multer server setup to the seven specialized interior planning modules and manual verification:

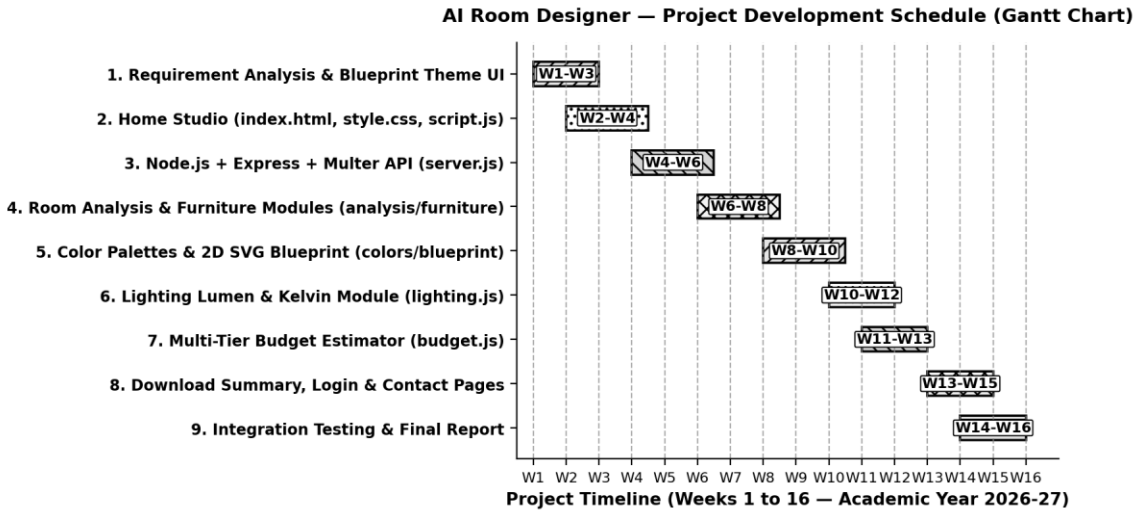


Figure 3.1: AI Room Designer — 16-Week Project Development Gantt Chart

■ Details of Hardware & Software Used

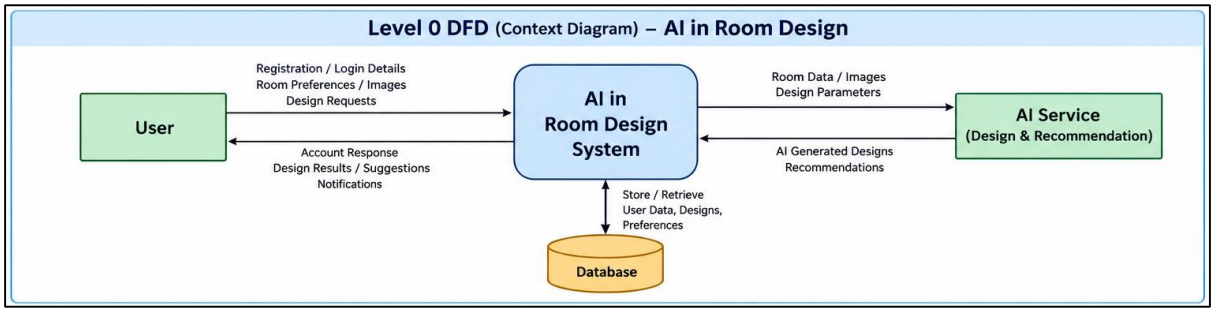
Hardware / Software Category	Specification & Purpose in AI-Room-Designer
Processor (CPU)	11th Gen Intel(R) Core(TM) i5-11320H @ 3.20 GHz (Clock Speed: 3.19 GHz)
Memory (RAM)	8.00 GB RAM (Minimum 4 GB required)
Disk Space	194 GB available storage (Minimum 20 GB for development tools and dependencies)
Operating System	Microsoft Windows 11 (64-bit development environment)
Code Editor / IDE	Visual Studio Code (Latest Version)
Backend Runtime & Framework	Node.js Runtime, Express.js (^5.1.0), Multer (Multipart file upload middleware)
Frontend Technologies	HTML5, CSS3 (Custom Blueprint Grid Variables, Flexbox, CSS Grid), Vanilla JavaScript (ES6+)
Graphics & Storage	Scalable Vector Graphics (Dynamic 2D SVG DOM), HTML5 FileReader API, Browser localStorage
Web Browser	Google Chrome / Microsoft Edge (for testing localStorage, SVG rendering, and print/download flows)

Data Flow Diagrams

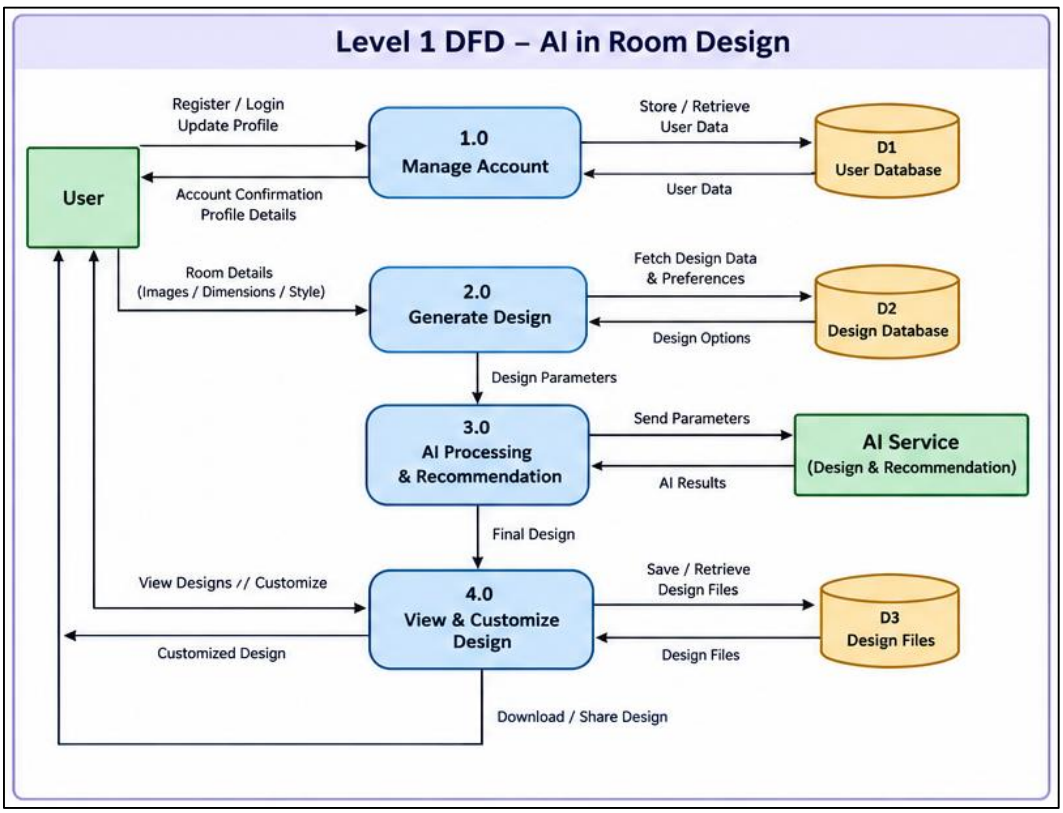
Data flow diagram

It shows how information enters and leaves the system, what changes the information and where information is stored. The purpose of a DFD is to show the scope and boundaries of a system as a whole. It may be used as a communications tool between a systems analyst and any person who plays a part in the system that acts as the starting point for redesigning a system.

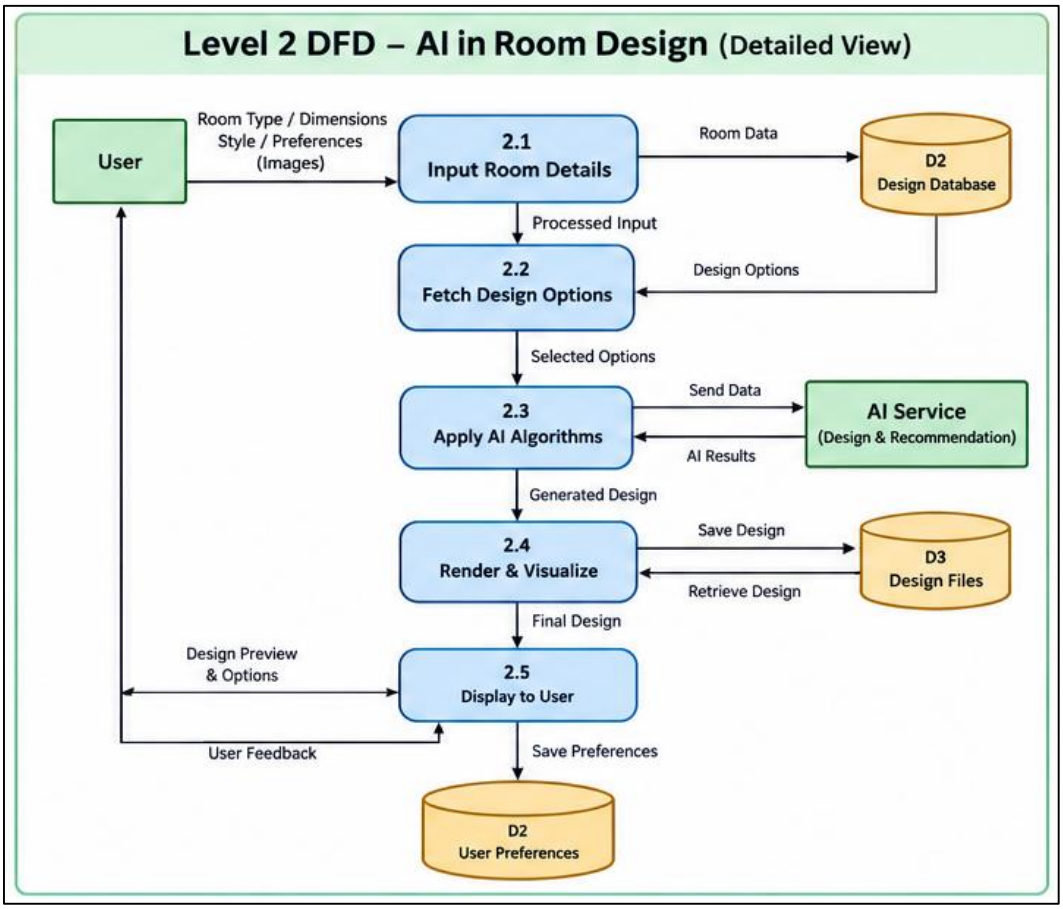
Data Flow Diagram — Level 0



First Level Data Flow Diagram



Second Level DFD

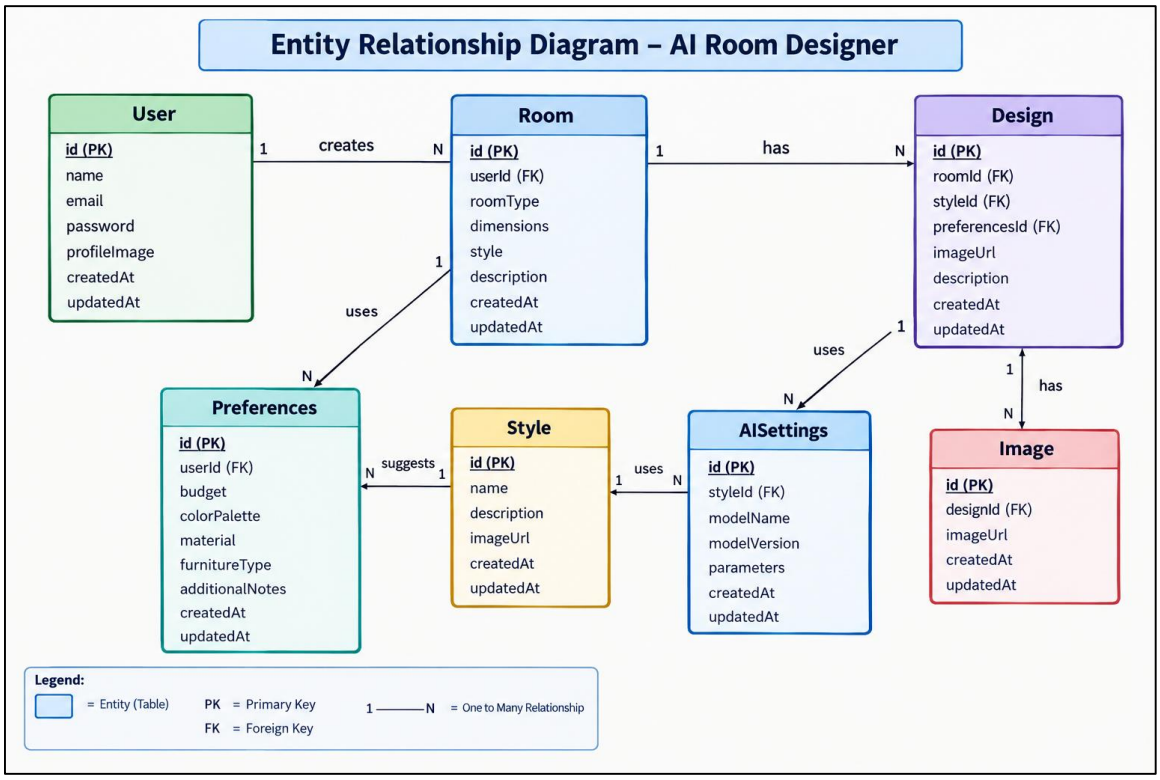


Entity Relationship diagram

ERD :

The Entity-Relationship Diagram (ERD) describes the data entities and relationships used in the AI Room Designer system. The diagram includes entities such as User, Room, Design, Preferences, Style, AI Settings, and Image. Rectangles represent the data entities, while lines connecting the rectangles show relationships between them. The final ERD should match the actual database schema and include only the models implemented in the project.

Entity Relationship Diagram



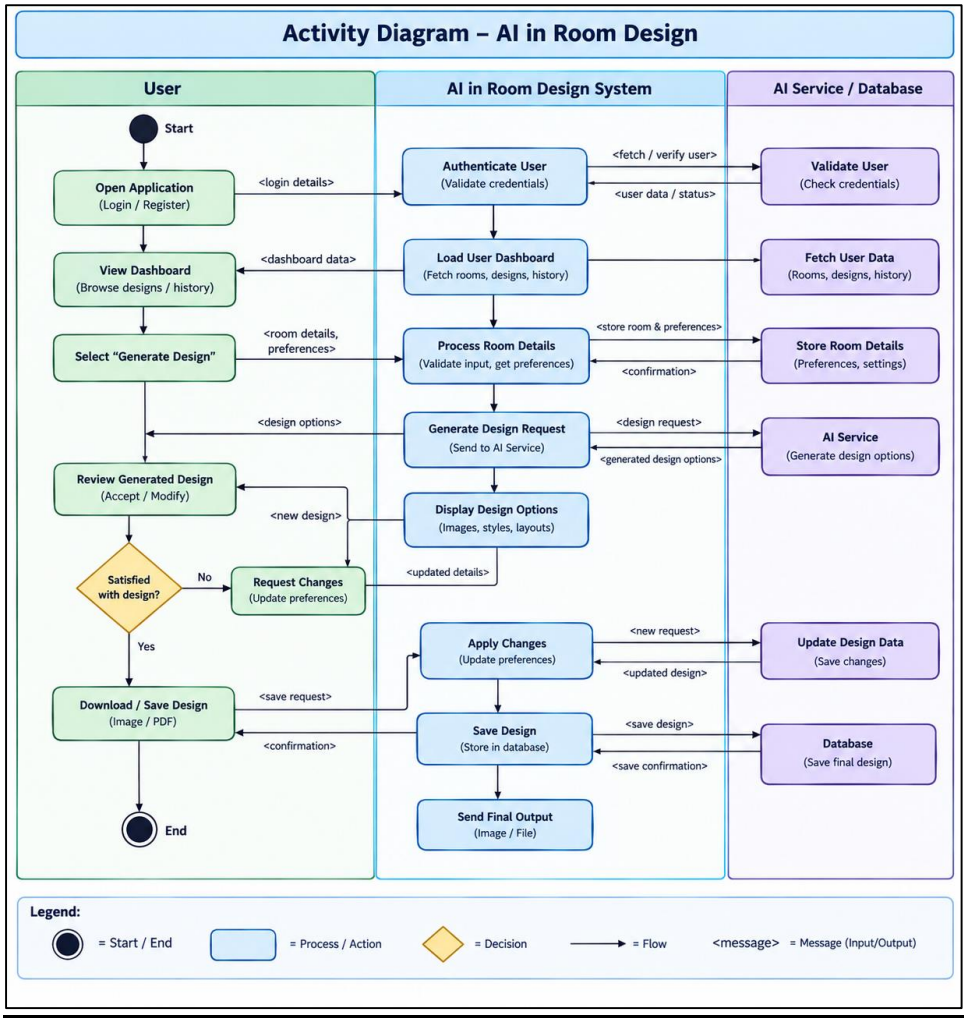
Activity Diagram

An activity diagram represents the sequence of actions in a workflow. For the AI in Room Design system, it shows a user opening the application, logging in or registering, entering room details and preferences, requesting an AI-generated design, reviewing the design, and saving or downloading the final result.

How to develop an activity diagram:

1. Describe messages from the external actor (User) to the system using message notation, such as login details, room dimensions, images, and design preferences.
2. Identify and add special conditions for input messages, including iteration and true/false decisions, such as whether the user is satisfied with the generated design or wants to make changes.
3. Identify and add output return messages, such as login confirmation, generated design options, updated designs, and final design download or save confirmation.

Activity Diagram



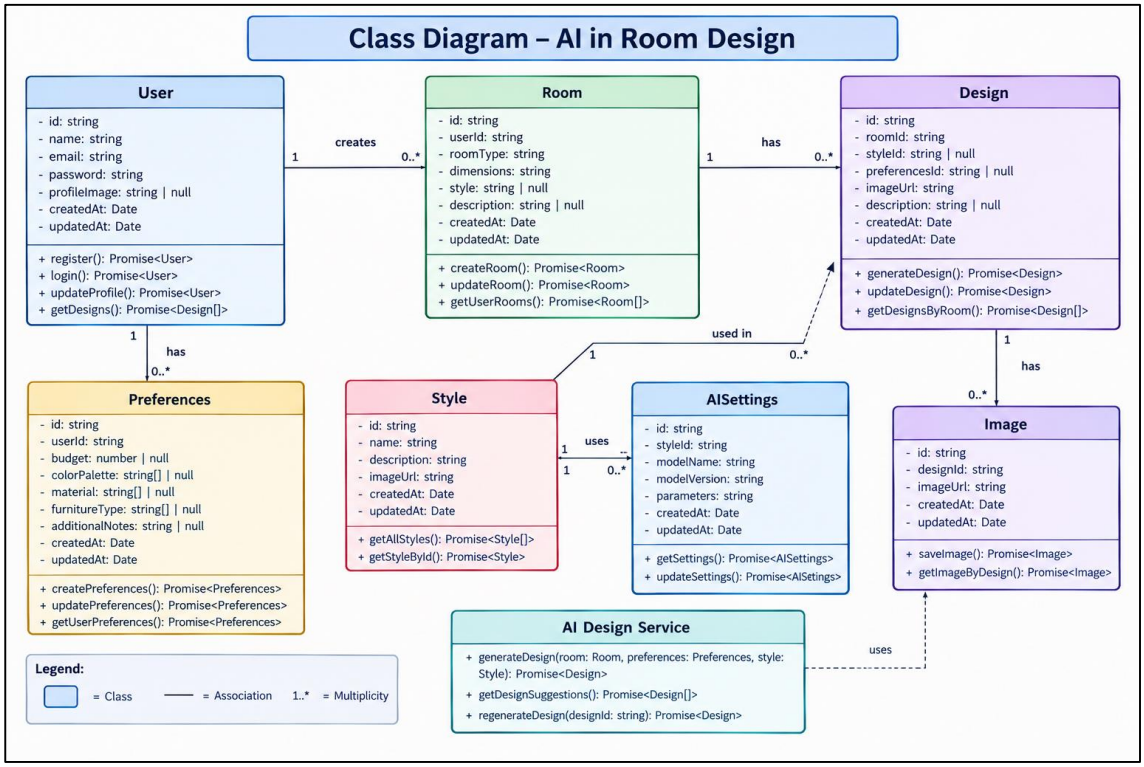
Class Diagram

A class diagram shows the static structure of an application, including its classes, attributes, methods, and relationships. For the AI in Room Design system, the diagram can represent classes such as User, Room, Design, Preferences, and AI Design Service, depending on the actual implementation.

A class describes the properties and operations of an object, while an association represents a relationship between classes. The diagram should be based on the actual application structure and implemented data models.

A class diagram describes the possible relationships among the components of the AI Room Design system rather than showing a single sequence of actions during runtime.

Class Diagram

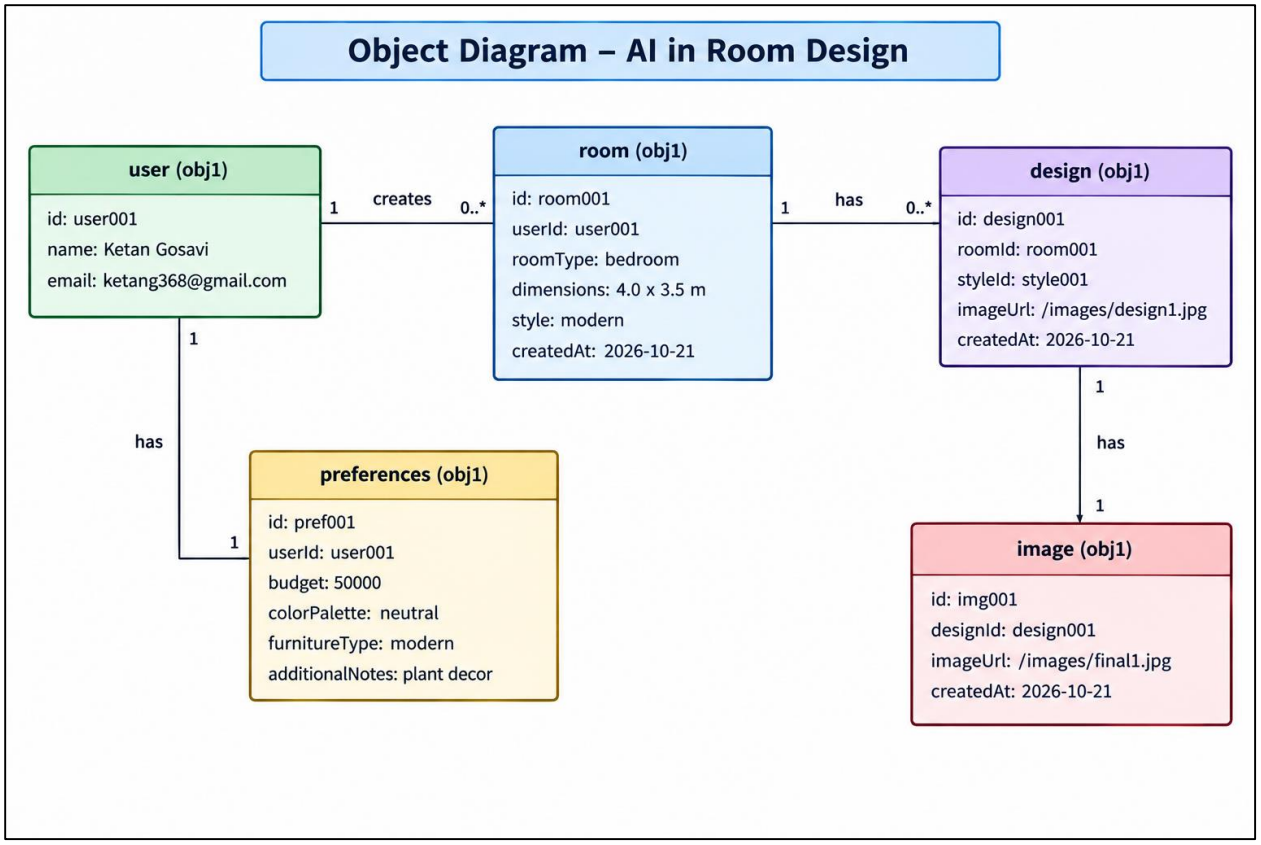


Object Diagram

An object diagram shows example instances of the data models and the relationships between them in the AI in Room Design system. It can include fictional examples of users, rooms, design preferences, and generated designs.

Object diagrams help explain how data is connected at a specific point in the workflow, such as a user linked to a bedroom, selected interior design preferences, and an AI-generated room design.

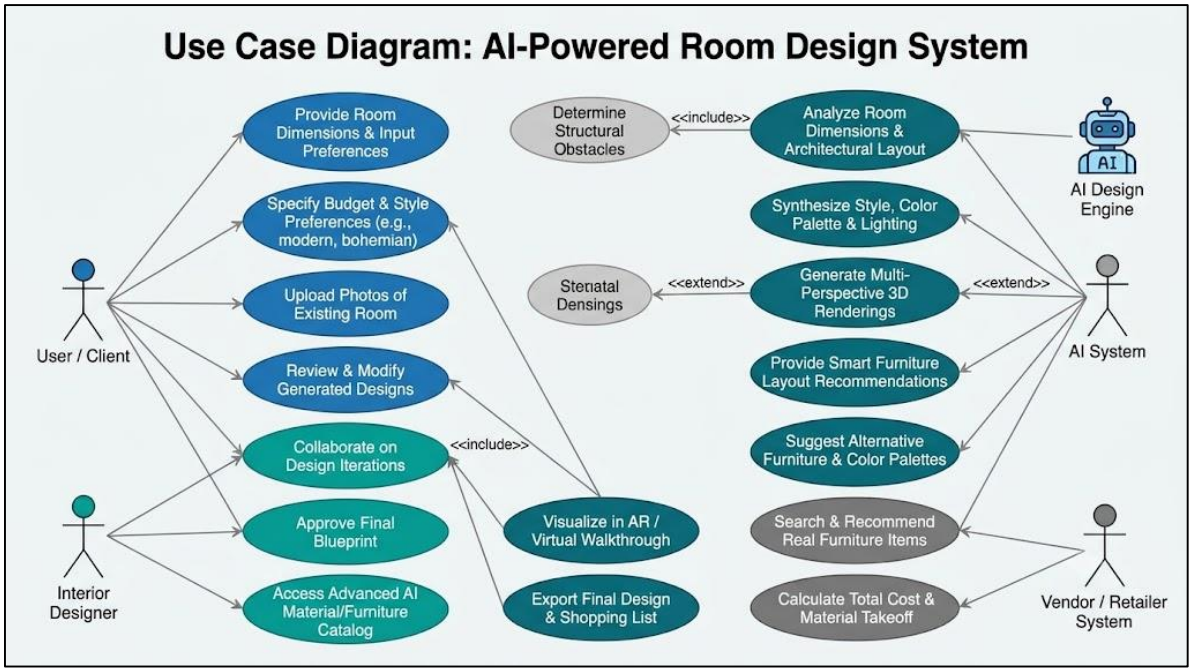
Object Diagram



Use Case Diagram

A use case describes a task that an actor performs with a system. The AI in Room Design use case diagram shows the user and the available functions, such as registering or logging in, entering room details, uploading room images, selecting design preferences, generating AI-based designs, viewing design recommendations, customizing designs, and saving or downloading the final design.

Use Case Diagram



Input and Output Screen Design

The following screenshots show the actual rendered screens of the AI Room Designer — Blueprint Edition web application across the home studio, all seven planning stages, login, and contact pages:

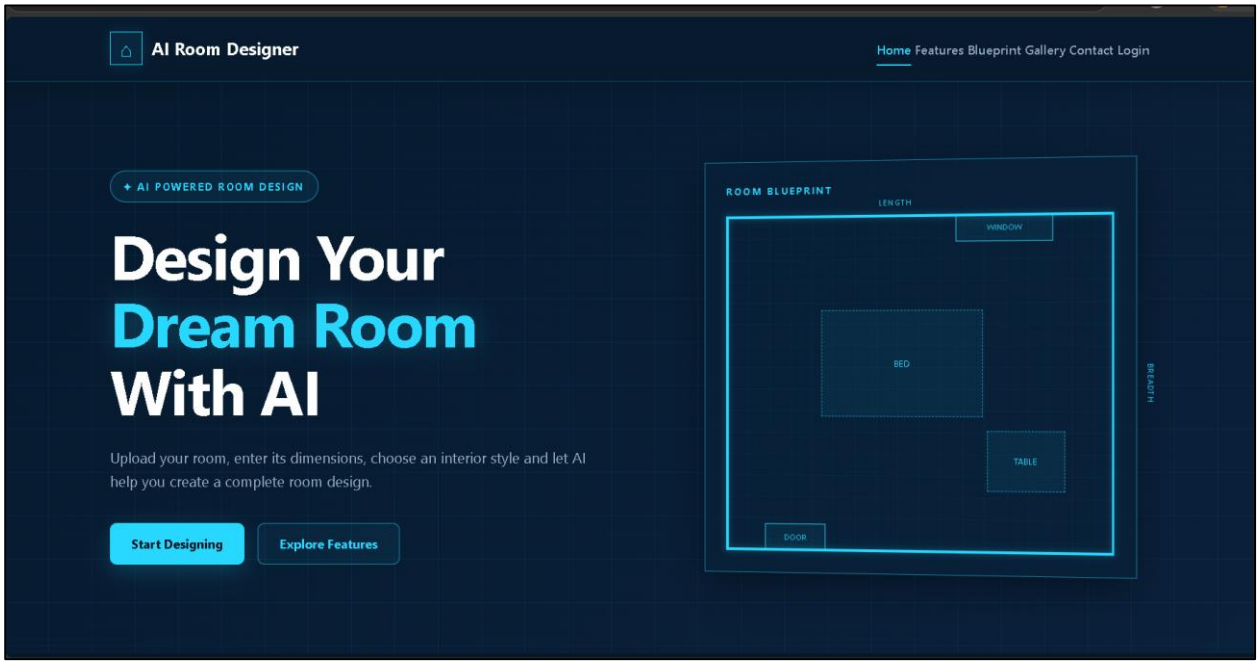


Figure 3.11: Home Page Hero Section & Quick Blueprint Preview (index.html)

Explanation: Displays the top navigation bar, 'Design Your Dream Room With AI' hero banner, summary statistics (5+ Design Styles, 2D Smart Blueprint, AI Room Analysis), and an interactive blueprint preview card.

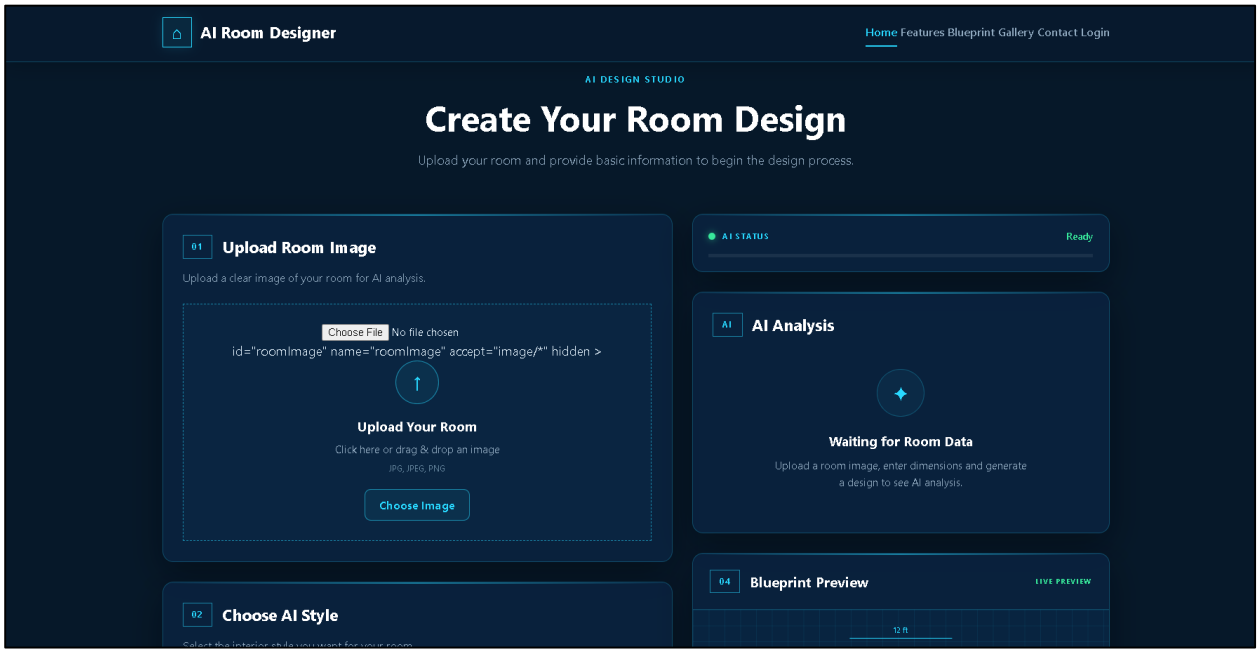


Figure 3.12: Main AI Design Studio — Image Upload, Room Dimensions & Style Selector (index.html)

Explanation: Allows uploading a room photo (JPG/PNG/WEBP), entering Length, Breadth, and Height (ft), choosing an interior style (Modern, Luxury, Minimal, Japanese, Scandinavian), and running the 5-step AI simulation.

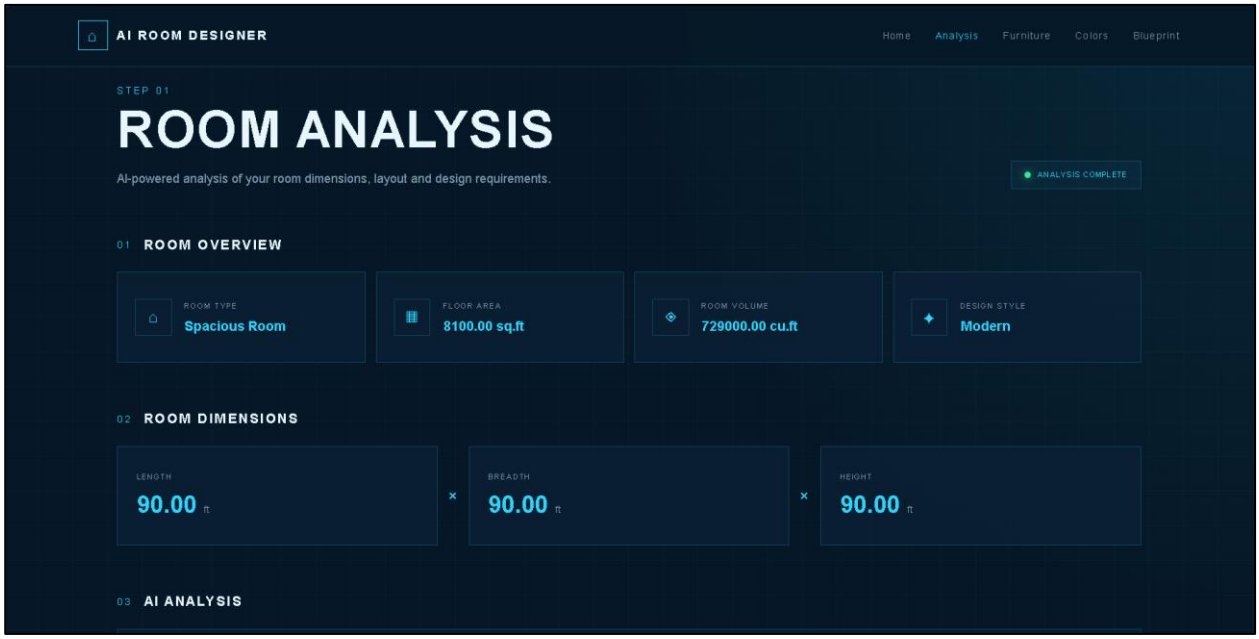


Figure 3.13: Stage 1 — AI Room Analysis Screen (pages/analysis.html)

Explanation: Reads aiRoomData from localStorage and presents the detected room classification (Large Room, 192 sq.ft, 1920 cu.ft), spatial assessment, and style-matched recommendations.

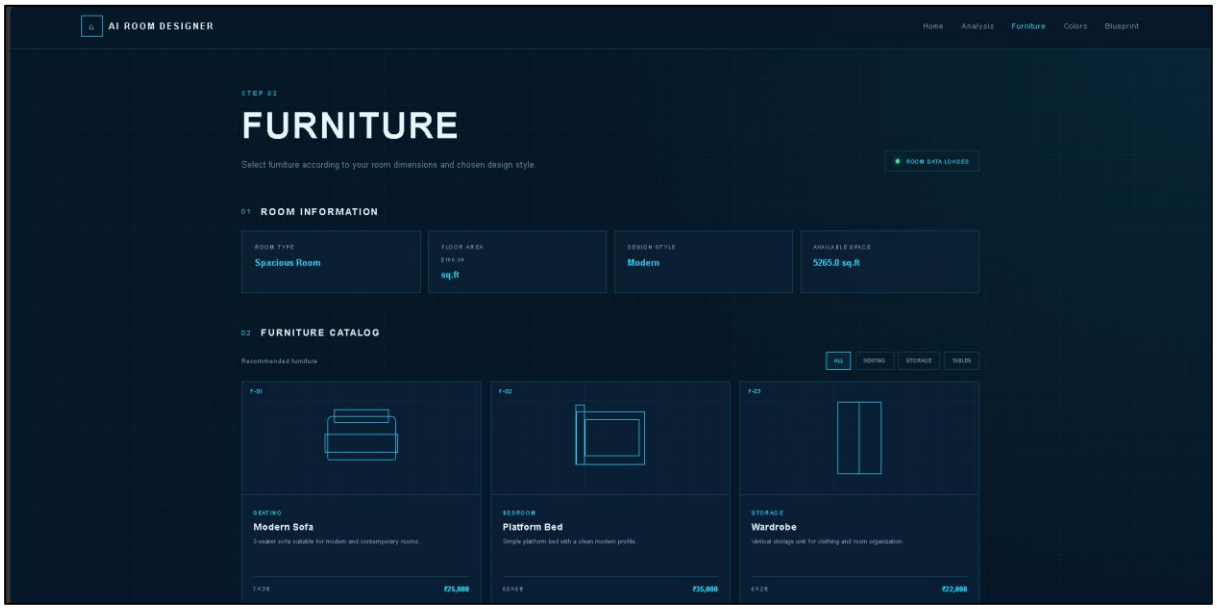


Figure 3.14: Stage 2 — Furniture Planning & 65% Usable Space Calculator (pages/furniture.html)

Explanation: Reserves 35% of floor area for movement clearance (showing 124.8 sq.ft usable space out of 192 sq.ft) and lets users filter and select furniture cards by category.

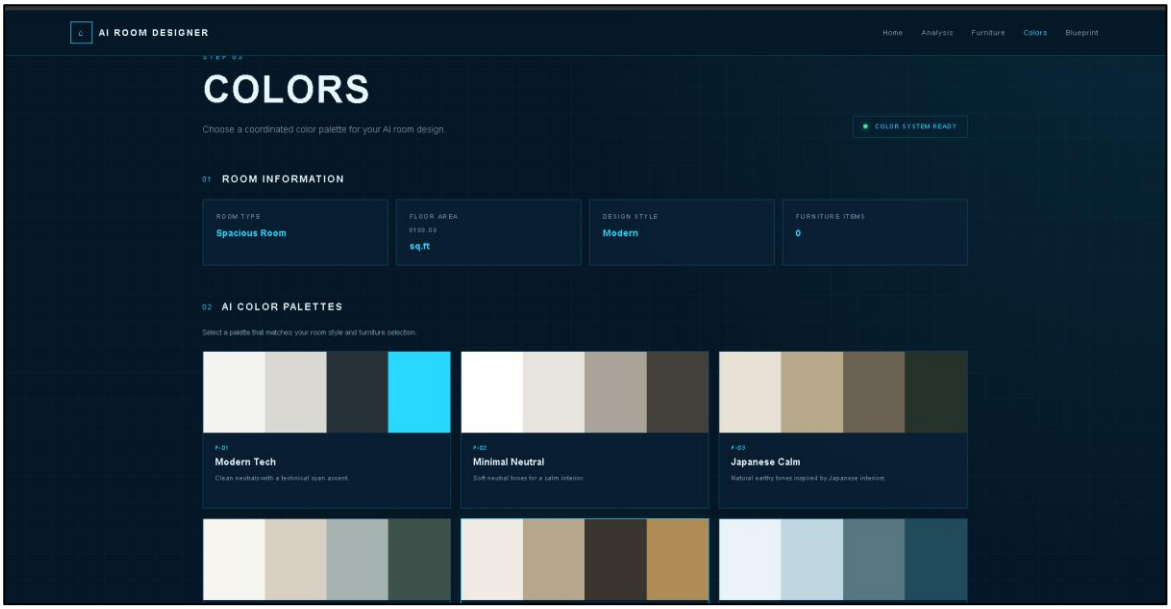


Figure 3.15: Stage 3 — 4-Channel Interior Color Palette Selector (pages/colors.html)

Explanation: Provides six curated color themes (Modern Tech, Minimal Neutral, Japanese Calm, Scandinavian, Luxury Warm, Cool Blue) with live Wall, Floor, Furniture, and Accent hex swatches.

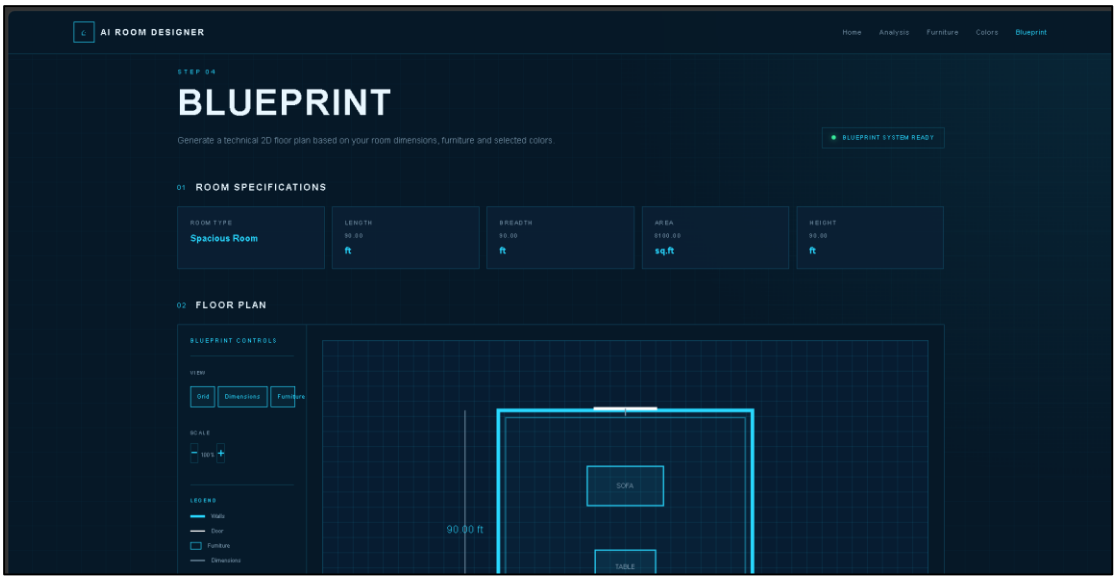


Figure 3.16: Stage 4 — Interactive 2D SVG Technical Floor Blueprint (pages/blueprint.html)

Explanation: Dynamically renders a scaled 1000×650 SVG floor plan (#blueprintSvg) complete with coordinate grid, room walls, door swing, window, furniture blocks (Sofa, Table, TV Unit, Plant), and zoom/export controls.

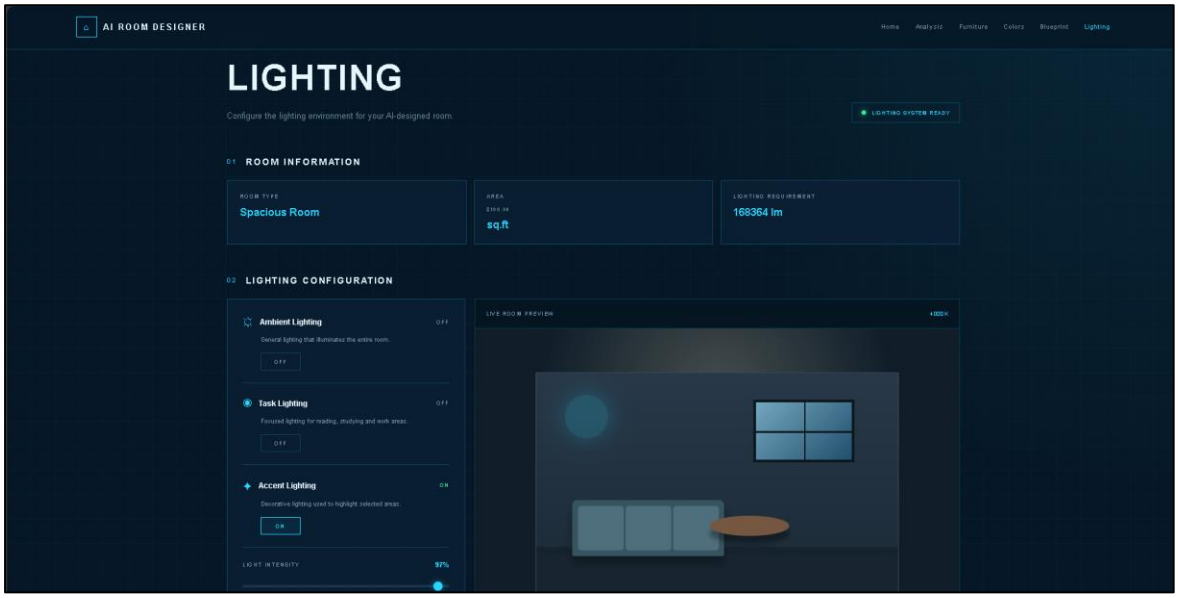


Figure 3.17: Stage 5 — Layered Lighting & Lumen Simulator (pages/lighting.html)

Explanation: Computes recommended lumens (Area × 15 scaled by intensity) and provides interactive toggles for Ambient, Task, and Accent lighting layers alongside Warm (2700K), Neutral (4000K), and Cool (5000K) modes.

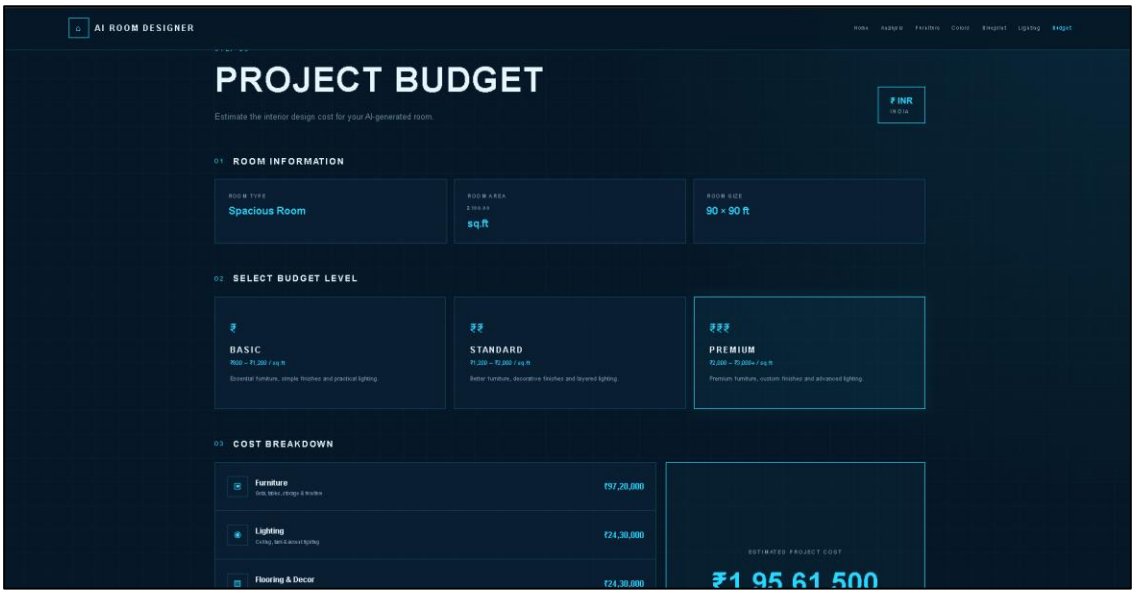


Figure 3.18: Stage 6 — Multi-Tier INR Renovation Budget Estimator (pages/budget.html)

Explanation: Calculates itemized costs across Basic, Standard, and Premium tiers for Furniture, Lighting, Decor, Painting, and Labour plus a 5% Contingency reserve (totaling INR 3,02,400 for a 192 sq.ft Standard room).

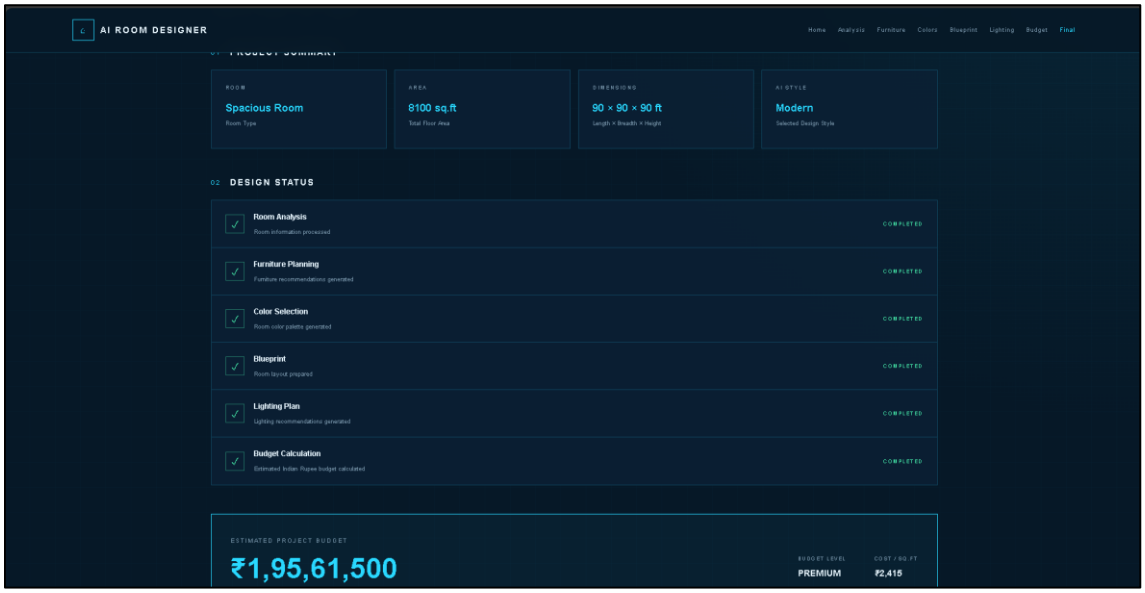


Figure 3.19: Stage 7 — Final Design Summary, Print & Report Download (pages/download.html)

Explanation: Consolidates room dimensions, chosen style, total INR budget, and cost per sq.ft., and provides buttons to download ai-room-design-summary.txt, print the report, or start a new design.

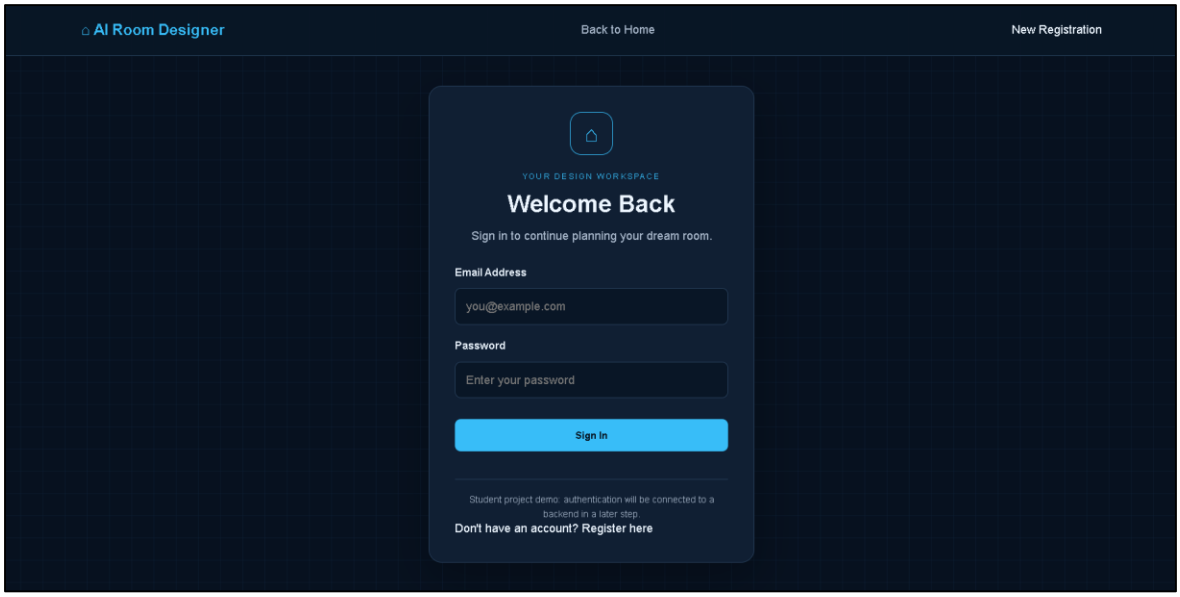


Figure 3.20: User Login Screen (pages/login.html)

Explanation: Provides email and password authentication fields with client-side validation in login.js.

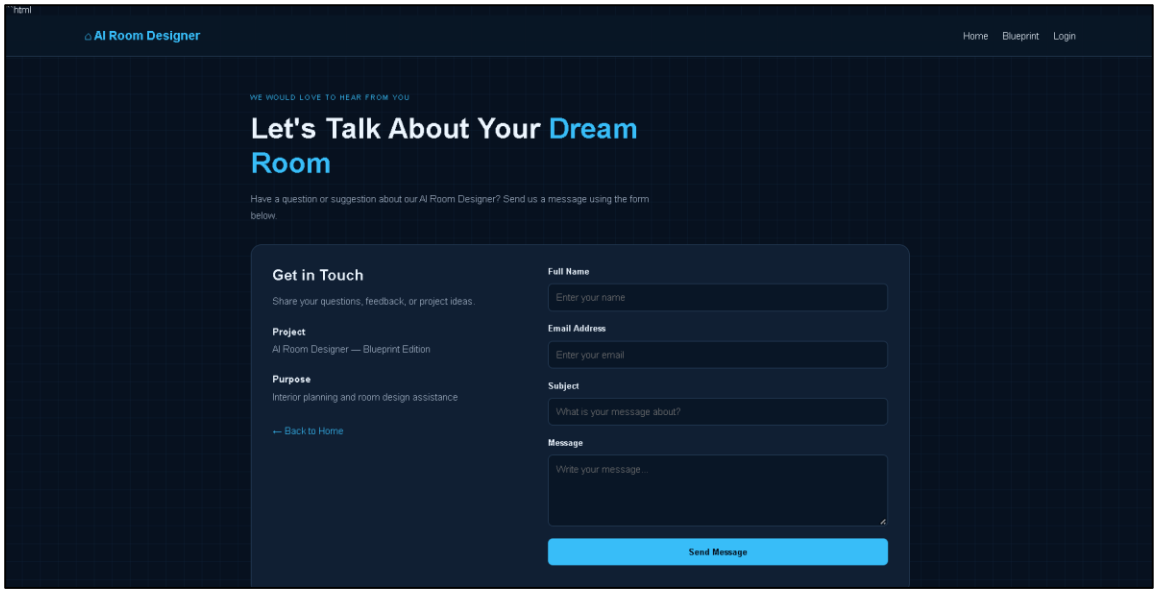


Figure 3.21: Contact & Feedback Screen (pages/contact.html)

Explanation: Allows users to submit project questions and feedback with field validation in contact.js.

■ Methodology Used for Testing & Test Reports

Manual Testing was performed across the Express backend endpoints (/api/status, /api/upload, /api/room/calculate) and all frontend modules by executing defined test cases with valid and invalid inputs:

Sr. No.	Test Case Description	Input / Action	Status	Expected Output	Actual Output
1	Backend Health Check (/api/status)	GET http://localhost:3000/api/status	Passed	Return JSON with success: true and status: 'online'.	Returned online status JSON.
2	Room Image Upload (/api/upload)	Upload valid PNG/JPG (< 5 MB)	Passed	File saved in /uploads/room-*.png and preview shown.	Saved in /uploads and preview rendered.
3	Invalid File Upload Filter	Upload non-image or > 5 MB file	Passed	Multer rejects upload with clear error message.	Error message displayed accurately.
4	Room Dimension Calculation	Length=16, Breadth=12, Height=10	Passed	Area=192 sq.ft, Volume=1920 cu.ft, Type='Large Room'.	Calculated 192 sq.ft & 1920 cu.ft.

Sr. No.	Test Case Description	Input / Action	Status	Expected Output	Actual Output
5	Zero / Negative Dimension Validation	Length=0 or Breadth=-4	Passed	Display error 'Room dimensions must be greater than zero.'	Validation blocked calculation.
6	5-Step AI Design Simulation	Select style & click Generate AI Design	Passed	Progress bar advances 55% -> 100% and renders recommendations.	Simulation completed at 100%.
7	Furniture 65% Usable Space Rule	Open furniture.html with 192 sq.ft room	Passed	Display 124.8 sq.ft available space (65% of 192).	124.8 sq.ft displayed; filters worked.
8	4-Channel Color Palette Selection	Select 'Japanese Calm' or 'Modern Tech'	Passed	Update Wall, Floor, Furniture, and Accent hex swatches.	Swatches and aiColorData updated.
9	2D SVG Blueprint Scaling & Zoom	Open blueprint.html & click Zoom/Grid	Passed	Render scaled 16×12 ft SVG plan with door, window & furniture.	SVG floor plan rendered and zoomed.
10	Lighting & Multi-Tier INR Budget	Select Standard tier on 192 sq.ft room	Passed	Compute Subtotal INR 2,88,000 + 5% Contingency = INR 3,02,400.	Exact INR totals and .txt export verified.

■ Code of the Project (Actual Source Files from AI-Room-Designer)

The following sections contain the actual implementation code from the AI-Room-Designer project, covering package.json, server.js, index.html, style.css, script.js, and the seven stage controllers in js/:

1. PACKAGE MANIFEST — package.json

```
{
  "name": "ai-room-designer", "version": "1.0.0", "description": "AI Room Designer
Blueprint Edition", "main": "server.js", "scripts": {
    "start": "node server.js"
  }, "dependencies": {
    "express": "^5.1.0"
  }
```

```
}
```

2. NODE.JS + EXPRESS + MULTER BACKEND — server.js

```
const express = require("express");
const path = require("path");
const multer = require("multer");
const fs = require("fs");
const app = express();
const PORT = 3000;
/* ========================================================== UPLOAD FOLDER
========================================================== */
const uploadDirectory = path.join( __dirname, "uploads"
);
/*
   Create uploads folder if it doesn't exist.
*/
if (!fs.existsSync(uploadDirectory)) {
    fs.mkdirSync( uploadDirectory,
        {
            recursive: true
        }
    );
}
/* ========================================================== MULTER STORAGE
========================================================== */
const storage = multer.diskStorage({
    destination: function (req, file, cb) {
        cb( null, uploadDirectory
        );
    }, filename: function (req, file, cb) {
        const extension = path.extname( file.originalname
            );
        const fileName = `room-${Date.now()}${extension}`;
        cb( null, fileName
        );
    }
});
/* ========================================================== IMAGE FILTER
========================================================== */
const imageFilter = function ( req, file, cb
) {
    const allowedTypes = [
        "image/jpeg", "image/png", "image/webp"
    ];
    if ( allowedTypes.includes( file.mimetype
        )
    ) {
```

```
        cb( null, true
        );
    }
    else {
        cb( new Error( "Only JPG, PNG and WEBP images are allowed."
            )
        );
    }
};
/* ============================================================ UPLOAD CONFIGURATION
============================================================ */
const upload = multer({
    storage: storage, fileFilter: imageFilter, limits: {
        fileSize:
            5 * 1024 * 1024
    }
});
/* ============================================================ MIDDLEWARE
============================================================ */
app.use( express.json()
);
app.use( express.urlencoded({
        extended: true
    })
);
/* ============================================================ STATIC FILES
============================================================ */
app.use( express.static( __dirname
    )
);
/* ============================================================ HOME PAGE
    // ... [Remaining helper functions in module] ...
        );
        console.log( `API status: http://localhost:${PORT}/api/status`
        );
        console.log( `Upload API: http://localhost:${PORT}/api/upload`
        );
        console.log( "========================================="
        );
        console.log("");
    }
);
```

3. MAIN STUDIO HTML — index.html (Header, Hero, Upload, Dimensions, Styles & Pipeline)

```
<!DOCTYPE html>
<html lang="en">
<head>
```

```
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description"
        content="AI Room Designer - Design your room using AI-powered room
analysis, furniture planning, colors, lighting and blueprint generation.">
    <title>AI Room Designer | Blueprint Edition</title>
    <!-- Main CSS -->
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <!-- ======================================================= HEADER / NAVIGATION
    ======================================================= -->
    <header class="header">
        <div class="container nav-container">
            <!-- Logo -->
            <a href="index.html" class="logo">
                <span class="logo-icon">⌂</span>
                <span>AI Room Designer</span>
            </a>
            <!-- Navigation -->
            <nav class="main-nav">
            <a href="#home" class="nav-link active">Home</a>
            <a href="#features" class="nav-link">Features</a>
            <a href="#blueprint" class="nav-link">Blueprint</a>
            <a href="#gallery" class="nav-link">Gallery</a>
            <a href="pages/contact.html" class="nav-link">Contact</a>
            <a href="pages/login.html" class="nav-link">Login</a>
            </nav>
            <!-- Mobile menu button -->
            <button class="menu-toggle" id="menuToggle" type="button">
                ☰
            </button>
        </div>
    </header>
    <!-- ======================================================= MAIN CONTENT
    ======================================================= -->
    <main>
        <!-- ================================================= HOME / HERO SECTION
        ================================================= -->
        <section id="home" class="hero-section">
            <div class="container hero-container">
                <!-- Hero Text -->
                <div class="hero-content">
                    <div class="hero-badge">
                        ✦ AI POWERED ROOM DESIGN
                    </div>
                    <h1>
                        Design Your
                        <span>Dream Room</span>
                        With AI
```

```
                    </h1>
                    <p class="hero-description">
                        Upload your room, enter its dimensions, choose an interior
style and let AI help
                        you create a complete room design.
                    </p>
                    <div class="hero-buttons">
                        <a href="#designer" class="btn btn-primary">
                            Start Designing
                        </a>
                        <a href="#features" class="btn btn-secondary">
                            Explore Features
                        </a>
                    </div>
                </div>
                <!-- Hero Blueprint Preview -->
                <div class="hero-blueprint">
                    <div class="blueprint-frame">
                        <div class="blueprint-label">
                            ROOM BLUEPRINT
                        </div>
                        <div class="blueprint-room">
                            <div class="blueprint-wall wall-top"></div>
                            <div class="blueprint-wall wall-right"></div>
    // ... [Remaining helper functions in module] ...
            </p>
            <a href="#home">
                Back to Top ↑
            </a>
        </div>
    </footer>
    <!-- Main JavaScript -->
    <script src="script.js"></script>
</body>
</html>
```

4. BLUEPRINT THEME STYLESHEET — style.css (Root Variables & Core Layout)

```
/* ========================================================== AI ROOM DESIGNER
   BLUEPRINT EDITION
   MAIN STYLESHEET
========================================================== */
/* ========================================================== 1. ROOT VARIABLES
========================================================== */
:root {
    --blueprint-dark: #071a2e;
    --blueprint-deep: #0a223d;
    --blueprint-blue: #102f50;
```

```
    --blueprint-light: #1c4d78;
    --cyan: #29d8ff;
    --cyan-light: #73e8ff;
    --white: #ffffff;
    --text: #e7f5ff;
    --muted: #91aac0;
    --border: rgba(41, 216, 255, 0.25);
    --border-strong: rgba(41, 216, 255, 0.55);
    --success: #38e89b;
    --warning: #ffc857;
    --danger: #ff647c;
    --card: rgba(10, 34, 61, 0.92);
    --shadow:
        0 15px 45px rgba(0, 0, 0, 0.35);
    --radius: 14px;
    --container: 1200px;
}
/* ========================================================== 2. RESET
========================================================== */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}
html {
    scroll-behavior: smooth;
}
body {
    font-family:
        "Segoe UI", Arial, sans-serif;
    background-color: var(--blueprint-dark);
    color: var(--text);
    line-height: 1.6;
    overflow-x: hidden;
}
body::before {
    content: "";
    position: fixed;
    inset: 0;
    pointer-events: none;
    z-index: -1;
    opacity: 0.35;
    background-image:
        linear-gradient( rgba(41, 216, 255, 0.08) 1px, transparent 1px
        ), linear-gradient( 90deg, rgba(41, 216, 255, 0.08) 1px, transparent 1px
        );
    background-size: 40px 40px;
}
img {
    max-width: 100%;
```

```
    display: block;
}
button, input {
    font-family: inherit;
}
button {
    cursor: pointer;
}
a {
    color: inherit;
    text-decoration: none;
}
/* ========================================================== 3. COMMON CONTAINER
========================================================== */
.container {
    // ... [Remaining helper functions in module] ...
    min-height: 650px;
    display: flex;
    align-items: center;
    padding: 90px 0;
    overflow: hidden;
    background:
        linear-gradient( 135deg, rgba(7, 26, 46, 0.97), rgba(10, 34, 61, 0.9)
        );
}
.hero-s
```

5. MAIN FRONTEND CONTROLLER — script.js (State, Upload, Dimensions, AI Simulation & API Fetch)

```
/* ========================================================== AI ROOM DESIGNER
   BLUEPRINT EDITION
   MAIN JAVASCRIPT
========================================================== */
/* ========================================================== 1. WAIT FOR PAGE TO
LOAD
========================================================== */
document.addEventListener("DOMContentLoaded", () => {
    console.log("AI Room Designer initialized.");
    initializeUpload();
    initializeDimensions();
    initializeStyleSelector();
    initializeDesignGenerator();
    initializeMobileMenu();
    initializeNavigation();
});
/* ========================================================== 2. GLOBAL PROJECT
STATE
```

```
========================================================== */
const roomData = {
    image: null, style: "modern", length: 0, breadth: 0, height: 0, area: 0,
volume: 0, roomType: "Not calculated", aiStatus: "Ready"
};
/* ========================================================== 3. IMAGE UPLOAD
========================================================== */
function initializeUpload() {
    const uploadArea = document.getElementById("uploadArea");
    const uploadButton = document.getElementById("uploadButton");
    const roomImage = document.getElementById("roomImage");
    const imagePreview = document.getElementById("imagePreview");
    const previewImage = document.getElementById("previewImage");
    const removeImage = document.getElementById("removeImage");
    if ( !uploadArea ||
        !uploadButton ||
        !roomImage
    ) {
        console.warn("Upload elements not found.");
        return;
    }
    /* ----------------------------------------
       Choose Image Button
    ---------------------------------------- */
    uploadButton.addEventListener("click", () => {
        roomImage.click();
    });
    /* ----------------------------------------
       Click Upload Area
    ---------------------------------------- */
    uploadArea.addEventListener("click", (event) => {
        if ( event.target === uploadButton ||
            event.target.closest("#uploadButton")
        ) {
            return;
        }
        roomImage.click();
    });
    /* ----------------------------------------
       File Selected
    ---------------------------------------- */
    roomImage.addEventListener("change", (event) => {
        const file = event.target.files[0];
        if (!file) {
            return;
        }
        handleImageUpload(file);
    });
    /* ----------------------------------------
       Remove Image
```

```
    ---------------------------------------- */
    if (removeImage) {
        removeImage.addEventListener("click", () => {
            roomImage.value = "";
            roomData.image = null;
            if (imagePreview) {
                imagePreview.hidden = true;
            }
            if (previewImage) {
                previewImage.src = "";
    // ... [Remaining helper functions in module] ...
        name: "Scandinavian", description:
            "A bright and comfortable interior using simple furniture and natural
materials.", colors: [
            "White", "Light Grey", "Soft Beige", "Light Wood"
        ], furniture: [
            "Fabric Sofa", "Wooden Coffee Table", "Open Shelf", "Simple TV Unit"
        ], lighting: [
            "Pendant Lights", "Warm LED Lights", "Natural Light"
        ]
    }
};
```

6. ROOM ANALYSIS MODULE — js/analysis.js

```
/* ========================================================== AI ROOM DESIGNER
   ANALYSIS PAGE JAVASCRIPT
========================================================== */
/* ========================================================== PAGE LOAD
========================================================== */
document.addEventListener( "DOMContentLoaded",
    () => {
        console.log( "Analysis page initialized."
        );
        loadRoomData();
        setupAnalysis();
    }
);
/* ========================================================== LOAD DATA FROM HOME
PAGE
========================================================== */
function loadRoomData() {
    /*
       Step 4 stores the room information
       in memory while index.html is open.
       localStorage allows us to keep the
       calculated information when moving
       between pages.
```

```
    */
    const savedData = localStorage.getItem( "aiRoomData"
        );
    if (!savedData) {
        console.log( "No saved room data found."
        );
        return;
    }
    try {
        const data = JSON.parse( savedData
            );
        updatePage( data
        );
    } catch (error) {
        console.error( "Could not load room data:", error
        );
    }
}
/* ========================================================== UPDATE PAGE
========================================================== */
function updatePage(data) {
    setText( "roomType", data.roomType || "Not calculated"
    );
    setText( "roomArea", formatNumber(data.area)
    );
    setText( "roomVolume", formatNumber(data.volume)
    );
    setText( "roomStyle", capitalize( data.style || "modern"
        )
    );
    setText( "lengthValue", formatNumber(data.length)
    );
    setText( "breadthValue", formatNumber(data.breadth)
    );
    setText( "heightValue", formatNumber(data.height)
    );
    generateRecommendations( data
    );
    generateAnalysis( data
    );
    setStatus( "ANALYSIS COMPLETE"
    );
}
/* ========================================================== AI ANALYSIS
========================================================== */
function generateAnalysis(data) {
    const content = document.getElementById( "analysisContent"
        );
    if (!content) {
        return;
```

```
    }
    const style = capitalize( data.style || "modern"
        );
    // ... [Remaining helper functions in module] ...
                    with a floor area of
                    <strong>${formatNumber(area)} sq.ft</strong>.
                </p>
                <p>
                    The selected design direction is
                    <strong>${style}</strong>.
                    ${spaceMessage}
                </p>
                <p>
                    The following pr
```

7. FURNITURE PLANNING MODULE — js/furniture.js

```
/* ========================================================== AI ROOM DESIGNER
   FURNITURE MODULE
========================================================== */
/* ========================================================== PROJECT STATE
========================================================== */
const furnitureState = {
    selected: [], totalCost: 0
};
/* ========================================================== PAGE LOAD
========================================================== */
document.addEventListener( "DOMContentLoaded",
    () => {
        console.log( "Furniture module initialized."
        );
        loadRoomData();
        initializeFilters();
        initializeFurnitureButtons();
        updateSelectedFurniture();
    }
);
/* ========================================================== LOAD ROOM DATA
========================================================== */
function loadRoomData() {
    const savedData = localStorage.getItem( "aiRoomData"
        );
    if (!savedData) {
        console.log( "No room data found."
        );
        return;
    }
    try {
```

```
        const data = JSON.parse( savedData
            );
        setText( "roomType", data.roomType ||
            "Not available"
        );
        setText( "roomArea", formatNumber(data.area)
        );
        setText( "roomStyle", capitalize( data.style ||
                "modern"
            )
        );
        calculateAvailableSpace( data
        );
    } catch (error) {
        console.error( "Room data error:", error
        );
    }
}
/* ========================================================== AVAILABLE SPACE
============================================================ */
function calculateAvailableSpace( data
) {
    const area = Number(data.area) || 0;
    let available;
    /*
       We reserve approximately 35%
       of floor area for movement.
    */
    available = area * 0.65;
    setText( "availableSpace",
        `${available.toFixed(1)} sq.ft`
    );
}
/* ========================================================== FILTER SYSTEM
============================================================ */
function initializeFilters() {
    const buttons = document.querySelectorAll( ".filter-button"
        );
    const cards = document.querySelectorAll( ".furniture-card"
        );
    buttons.forEach( (button) => {
            button.addEventListener( "click",
                () => {
                    buttons.forEach( item =>
    // ... [Remaining helper functions in module] ...
                            }
                        }
                    );
                }
            );
```

```
        }
    );
}
/* ========================================================== ADD / REMOVE
FURNITURE
=======================
```

8. 4-CHANNEL COLOR PALETTE MODULE — js/colors.js

```
/* ========================================================== AI ROOM DESIGNER
   COLORS MODULE
============================================================ */
/* ========================================================== COLOR PALETTE DATA
============================================================ */
const colorPalettes = {
    modern: {
        name: "Modern Tech", description:
            "Clean neutrals with a technical cyan accent.", wall: "#F4F4F0",
floor: "#D9D9D2", furniture: "#263238", accent: "#29D8FF"
    }, minimal: {
        name: "Minimal Neutral", description:
            "Soft neutral tones for a calm interior.", wall: "#FFFFFF", floor:
"#E8E5DF", furniture: "#A9A39A", accent: "#44403C"
    }, japanese: {
        name: "Japanese Calm", description:
            "Natural earthy tones inspired by Japanese interiors.", wall:
"#E8E1D5", floor: "#B8A98A", furniture: "#6B6254", accent: "#26332D"
    }, scandinavian: {
        name: "Scandinavian", description:
            "Light natural colors with subtle green accents.", wall: "#F7F5EF",
floor: "#D8D0C3", furniture: "#A7B3B0", accent: "#3E514A"
    }, luxury: {
        name: "Luxury Warm", description:
            "Warm neutrals with elegant metallic-inspired accents.", wall:
"#F1ECE3", floor: "#B7A78C", furniture: "#3B3530", accent: "#B08D57"
    }, cool: {
        name: "Cool Blue", description:
            "Fresh blue-grey tones for a modern atmosphere.", wall: "#EAF4F8",
floor: "#BFD7E0", furniture: "#587783", accent: "#214B5A"
    }
};
/* ========================================================== CURRENT STATE
============================================================ */
let selectedPalette = "modern";
/* ========================================================== PAGE LOAD
============================================================ */
document.addEventListener( "DOMContentLoaded",
    () => {
```

```
        console.log( "Colors module initialized."
        );
        loadRoomData();
        loadSavedPalette();
        initializePaletteSelection();
        updatePalettePreview();
    }
);
/* ========================================================== LOAD ROOM DATA
========================================================== */
function loadRoomData() {
    const roomData = localStorage.getItem( "aiRoomData"
        );
    if (roomData) {
        try {
            const data = JSON.parse(roomData);
            setText( "roomType", data.roomType ||
                "Not available"
            );
            setText( "roomArea", formatNumber(data.area)
            );
            setText( "roomStyle", capitalize( data.style ||
                    "modern"
                )
            );
        } catch (error) {
            console.error( "Room data error:", error
            );
        }
    }
    loadFurnitureCount();
}
/* ========================================================== LOAD FURNITURE COUNT
========================================================== */
function loadFurnitureCount() {
    const furnitureData = localStorage.getItem( "aiFurnitureData"
        );
    if (!furnitureData) {
        setText( "furnitureCount", "0"
        );
        return;
    }
    // ... [Remaining helper functions in module] ...
        const data = JSON.parse( furnitureData
            );
        const count = Array.isArray(data.selected)
                ? data.selected.length
                : 0;
        setText( "furnitureCount", count
        );
```

```
    } catch (error) {
        console.error( "Furniture data error:", error
        )
```

9. DYNAMIC 2D SVG BLUEPRINT GENERATOR — js/blueprint.js

```
/* ========================================================== AI ROOM DESIGNER
   BLUEPRINT MODULE
========================================================== */
/* ============================================================ GLOBAL DATA
============================================================ */
let roomData = {
    length: 12, breadth: 10, height: 10, area: 120, roomType: "Living Room"
};
let colorData = {
    palette: "modern", wall: "#F4F4F0", floor: "#D9D9D2", furniture: "#263238",
accent: "#29D8FF"
};
let zoomLevel = 1;
/* ============================================================ PAGE LOAD
============================================================ */
document.addEventListener( "DOMContentLoaded",
    () => {
        console.log( "Blueprint module initialized."
        );
        loadRoomData();
        loadColorData();
        updateRoomInformation();
        generateBlueprint();
        initializeControls();
    }
);
/* ========================================================== LOAD ROOM DATA
========================================================== */
function loadRoomData() {
    const savedRoomData = localStorage.getItem( "aiRoomData"
        );
    if (!savedRoomData) {
        console.log( "No saved room data found. Using demo dimensions."
        );
        return;
    }
    try {
        const data = JSON.parse( savedRoomData
            );
        roomData = {
            ...roomData,
            ...data
```

```
        };
        roomData.length = Number( data.length ||
                data.roomLength ||
                roomData.length
            );
        roomData.breadth = Number( data.breadth ||
                data.roomBreadth ||
                roomData.breadth
            );
        roomData.height = Number( data.height ||
                data.roomHeight ||
                roomData.height
            );
        roomData.area = Number( data.area ||
                roomData.length *
                roomData.breadth
            );
    } catch (error) {
        console.error( "Room data error:", error
        );
    }
}
/* ========================================================== LOAD COLOR DATA
============================================================ */
function loadColorData() {
    const savedColorData = localStorage.getItem( "aiColorData"
        );
    if (!savedColorData) {
        return;
    }
    try {
        const data = JSON.parse( savedColorData
            );
        colorData = {
    // ... [Remaining helper functions in module] ...
    const maxRoomHeight = 420;
    const scaleX = maxRoomWidth /
        roomData.length;
    const scaleY = maxRoomHeight /
        roomData.breadth;
    const scale = Math.min( scaleX, scaleY
        );
    const roomWidth = roomData.length *
        scale;
    const roomHeight
```

10. LAYERED LIGHTING & LUMEN MODULE — js/lighting.js

```
/* ========================================================== AI ROOM DESIGNER
   LIGHTING MODULE
========================================================== */
/* ========================================================== DEFAULT DATA
========================================================== */
let roomData = {
    roomType: "Living Room", length: 12, breadth: 10, height: 10, area: 120
};
let lightingData = {
    ambient: true, task: true, accent: true, intensity: 70, temperature:
"neutral", temperatureValue: 4000
};
/* ========================================================== PAGE LOAD
========================================================== */
document.addEventListener( "DOMContentLoaded",
    () => {
        console.log( "Lighting module initialized."
        );
        loadRoomData();
        loadLightingData();
        updateRoomSummary();
        initializeLightingControls();
        updateLightingPreview();
        updateRecommendation();
    }
);
/* ========================================================== LOAD ROOM DATA
========================================================== */
function loadRoomData() {
    const savedData = localStorage.getItem( "aiRoomData"
        );
    if (!savedData) {
        return;
    }
    try {
        const data = JSON.parse( savedData
            );
        roomData = {
            ...roomData,
            ...data
        };
        roomData.length = Number( data.length ||
                data.roomLength ||
                roomData.length
            );
        roomData.breadth = Number( data.breadth ||
                data.roomBreadth ||
                roomData.breadth
```

```
            );
        roomData.height = Number( data.height ||
                data.roomHeight ||
                roomData.height
            );
        roomData.area = roomData.length *
            roomData.breadth;
    } catch (error) {
        console.error( "Room data could not be loaded:", error
        );
    }
}
/* ========================================================== LOAD LIGHTING DATA
========================================================== */
function loadLightingData() {
    const savedData = localStorage.getItem( "aiLightingData"
        );
    if (!savedData) {
        return;
    }
    try {
        lightingData = {
            ...lightingData,
            ...JSON.parse( savedData
            )
        };
    } catch (error) {
        console.error( "Lighting data could not be loaded:", error
    // ... [Remaining helper functions in module] ...
        ) +
        " lm"
    );
}
/* ========================================================== INITIALIZE CONTROLS
========================================================== */
function initializeLightingControls() {
    /*
       Ambient
    */
```

11. MULTI-TIER INR BUDGET MODULE — js/budget.js

```
/* ========================================================== AI ROOM DESIGNER
   BUDGET MODULE
========================================================== */
/* ========================================================== ROOM DATA
========================================================== */
let roomData = {
```

```
    roomType: "Living Room", length: 12, breadth: 10, height: 10, area: 120
};
/* ========================================================= BUDGET DATA
========================================================= */
let budgetData = {
    level: "standard", furniture: 0, lighting: 0, decor: 0, painting: 0, labour:
0, contingency: 0, total: 0
};
/* ========================================================= BUDGET RATES
========================================================= */
/*
   These are simplified project estimates.
   They are NOT vendor quotations.
   The rates are intentionally kept as
   approximate planning values for the
   TYCS mini-project.
*/
const budgetRates = {
    basic: {
        furniture: 500, lighting: 120, decor: 100, painting: 80, labour: 150
    }, standard: {
        furniture: 800, lighting: 180, decor: 180, painting: 120, labour: 220
    }, premium: {
        furniture: 1200, lighting: 300, decor: 300, painting: 180, labour: 320
    }
};
/* ========================================================= PAGE LOAD
========================================================= */
document.addEventListener( "DOMContentLoaded",
    () => {
        console.log( "Budget module initialized."
        );
        loadRoomData();
        loadBudgetData();
        updateRoomInformation();
        initializeBudgetOptions();
        calculateBudget();
        updateRecommendation();
    }
);
/* ========================================================= LOAD ROOM DATA
========================================================= */
function loadRoomData() {
    const savedData = localStorage.getItem( "aiRoomData"
        );
    if (!savedData) {
        return;
    }
    try {
        const data = JSON.parse( savedData
```

```
            );
        roomData = {
            ...roomData,
            ...data
        };
        roomData.length = Number( data.length ||
                data.roomLength ||
                roomData.length
            );
        roomData.breadth = Number( data.breadth ||
                data.roomBreadth ||
                roomData.breadth
            );
        roomData.height = Number( data.height ||
                data.roomHeight ||
                roomData.height
            );
        roomData.area = roomData.length *
            roomData.breadth;
    } catch (error) {
    // ... [Remaining helper functions in module] ...
    setText( "roomArea", area.toFixed(2)
    );
    setText( "roomDimensions",
        `${roomData.length} × ${roomData.breadth} ft`
    );
}
/* ========================================================== INITIALIZE BUDGET
OPTIONS
========================================================== */
function initializeBudgetOptions() {
    const options =
```

12. FINAL SUMMARY & REPORT DOWNLOAD MODULE — js/download.js

```
/* ========================================================== AI ROOM DESIGNER
   DOWNLOAD / FINAL MODULE
========================================================== */
/* ========================================================== DEFAULT DATA
========================================================== */
let roomData = {
    roomType: "Living Room", length: 12, breadth: 10, height: 10, area: 120,
style: "Modern"
};
let budgetData = {
    level: "standard", total: 0, contingency: 0
};
/* ========================================================== PAGE LOAD
```

```
========================================================== */
document.addEventListener( "DOMContentLoaded",
    () => {
        console.log( "Final download module initialized."
        );
        loadRoomData();
        loadBudgetData();
        updateSummary();
        initializeButtons();
    }
);
/* ========================================================== LOAD ROOM DATA
========================================================== */
function loadRoomData() {
    const savedData = localStorage.getItem( "aiRoomData"
        );
    if (!savedData) {
        return;
    }
    try {
        const data = JSON.parse( savedData
            );
        roomData = {
            ...roomData,
            ...data
        };
        roomData.length = Number( data.length ||
                data.roomLength ||
                roomData.length
            );
        roomData.breadth = Number( data.breadth ||
                data.roomBreadth ||
                roomData.breadth
            );
        roomData.height = Number( data.height ||
                data.roomHeight ||
                roomData.height
            );
        roomData.area = roomData.length *
            roomData.breadth;
        /*
           Different pages may save the style
           under different names.
        */
        roomData.style = data.style ||
            data.roomStyle ||
            data.aiStyle ||
            roomData.style;
    }
    catch (error) {
```

```
        console.error( "Unable to load room data:", error
        );
    }
}
/* ========================================================== LOAD BUDGET DATA
========================================================== */
function loadBudgetData() {
    const savedData = localStorage.getItem( "aiBudgetData"
        );
    if (!savedData) {
        return;
    }
    try {
    // ... [Remaining helper functions in module] ...
    );
    /*
       AI recommendation
    */
    updateRecommendation();
}
/* ========================================================== AI RECOMMENDATION
========================================================== */
function updateRecommendation() {
    let title = "Balanced Modern Int
```

13. LOGIN & CONTACT VALIDATION MODULES — js/login.js & js/contact.js

```
// --- js/login.js ---
const loginForm = document.getElementById("loginForm");
const loginStatus = document.getElementById("loginStatus");
loginForm.addEventListener("submit", function (event) {
    event.preventDefault();
    const email = document.getElementById("loginEmail").value.trim();
    const password = document.getElementById("loginPassword").value;
    if (!email || !password) {
        loginStatus.textContent = "Please enter your email and password.";
        return;
    }
    loginStatus.textContent = "Login page is working. Account verification will be
connected to the backend later.";
});
// --- js/contact.js ---
const contactForm = document.getElementById("contactForm");
const contactStatus = document.getElementById("contactStatus");
contactForm.addEventListener("submit", function (event) {
    event.preventDefault();
    const name = document.getElementById("contactName").value.trim();
    const email = document.getElementById("contactEmail").value.trim();
```

```
    const subject = document.getElementById("contactSubject").value.trim();
    const message = document.getElementById("contactMessage").value.trim();
    if (!name || !email || !subject || !message) {
        contactStatus.textContent = "Please complete all fields.";
        return;
    }
    contactStatus.textContent = "Form validated successfully. Message sending will
be connected to the backend later.";
    contactForm.reset();
});
```

CHAPTER IV

Annexure

CHAPTER IV

DATABASE & APPLICATION DATA STRUCTURES

AI Room Designer — Blueprint Edition uses structured JavaScript state objects persisted in browser localStorage alongside Express REST API payloads. The tables below document the fixed enumerations, budget rate matrices, color palette definitions, and core state entities implemented in the project.

4.1 Fixed Choice Groups and Allowed Values

Parameter / Choice Group	Allowed Values in AI-Room-Designer
Interior Styles (roomData.style)	modern, luxury, minimal, japanese, scandinavian
Room Size Classifications (roomData.roomType)	Small Room (<80 sq.ft), Medium Room (80–149 sq.ft), Large Room (150–249 sq.ft), Spacious Room (>=250 sq.ft)
Color Palettes (colors.js)	modern (Modern Tech), minimal (Minimal Neutral), japanese (Japanese Calm), scandinavian (Scandinavian), luxury (Luxury Warm), cool (Cool Blue)
Lighting Layers & Kelvin Modes (lighting.js)	Layers: ambient, task, accent (Boolean) | Temperatures: warm (2700K), neutral (4000K), cool (5000K)
Budget Tiers (budget.js)	basic (INR 950/sq.ft base), standard (INR 1,500/sq.ft base), premium (INR 2,300/sq.ft base) + 5% Contingency
Allowed Upload MIME Types (server.js)	image/jpeg (.jpg, .jpeg), image/png (.png), image/webp (.webp) — Max File Size: 5 MB

4.2 Multi-Tier Renovation Budget Rate Matrix (budget.js)

Budget Tier	Furniture (/sq.ft)	Lighting (/sq.ft)	Decor (/sq.ft)	Painting (/sq.ft)	Labour (/sq.ft)	Contingency
Basic Tier	INR 500	INR 120	INR 100	INR 80	INR 150	5% of Subtotal
Standard Tier	INR 800	INR 180	INR 180	INR 120	INR 220	5% of Subtotal
Premium Tier	INR 1,200	INR 300	INR 300	INR 180	INR 320	5% of Subtotal

4.3 Core Application Data Entities & localStorage Keys

State Key / Entity	Module Owner	Attributes Stored	Design Purpose
aiRoomData	script.js, analysis.js, blueprint.js	image, style, length, breadth, height, area, volume, roomType, aiStatus	Primary spatial and style state shared across all 7 stage pages.
uploadedRoomImage	script.js, server.js (/api/upload)	originalName, fileName, size, path (/uploads/room-*.png)	Tracks Multer-uploaded room photograph metadata.
furnitureState	js/furniture.js	selected (Array of items), totalCost, availableSpace (area × 0.65)	Enforces 35% movement clearance and tracks selected furniture cost.
aiColorData	js/colors.js, js/blueprint.js	palette, wall, floor, furniture, accent (Hex color codes)	Synchronizes selected 4-channel color swatches with the 2D blueprint.
aiLightingData	js/lighting.js	ambient, task, accent, intensity (%), temperature, temperatureValue (K)	Controls lighting preview layers and lumen calculation.
aiBudgetData	js/budget.js, js/download.js	level, furniture, lighting, decor, painting, labour, contingency, total	Stores itemized INR renovation estimate for final report export.

CHAPTER V

Conclusion

CHAPTER V
CONCLUSION

AI Room Designer — Blueprint Edition successfully centralizes interior room planning into a cohesive seven-stage web application. By combining room photograph upload, automated Length × Breadth × Height spatial calculations, five curated interior style profiles, furniture clearance calculation, four-channel color palettes, dynamic 2D SVG blueprint generation, layered lighting simulation, and multi-tier INR budget estimation, the system eliminates the need to switch between disconnected planning tools.

The integration of a lightweight Node.js + Express + Multer backend (server.js) with modular frontend controllers and browser localStorage state persistence ensures fast responsiveness, clear validation feedback, and reliable cross-page data continuity. Users can move effortlessly from initial room measurements to a printable and downloadable design specification.

CHAPTER VI

Future Enhancement

CHAPTER VI

FUTURE ENHANCEMENT

While AI Room Designer — Blueprint Edition provides a complete end-to-end room planning workflow, future versions of the system can be expanded with the following enhancements:

- **1. Live Generative AI Image-to-Image Rendering:** Connect uploaded room photographs in /uploads to a cloud diffusion or Gemini Vision endpoint to render photorealistic restyled room images.
- **2. Interactive Drag-and-Drop SVG Furniture Placement:** Extend blueprint.js so users can drag, rotate, and snap individual furniture blocks directly inside the 2D SVG floor plan.
- **3. Persistent Cloud Database Integration (PostgreSQL / MongoDB):** Connect login.js and contact.js to persistent backend database tables so users can save multiple room projects across devices.
- **4. 3D WebGL / Three.js Room Walkthrough:** Extrude the 2D SVG floor plan into an interactive 3D room view with live wall colors and directional lighting.
- **5. PDF Blueprint & Quotation Export:** Enhance download.js to export a styled PDF blueprint and itemized contractor quotation sheet alongside the .txt summary.
- **6. E-Commerce Furniture Catalog Links:** Link furniture cards in furniture.html to real-time product listings from online furniture stores.

CHAPTER VII

References & Bibliography

CHAPTER VII

REFERENCES AND BIBLIOGRAPHY

The following technical documentation and web resources were used as references during the development of AI Room Designer — Blueprint Edition:

Reference Source	URL / Documentation Link	Technical Area Covered
MDN Web Docs (HTML5, CSS3, JavaScript, SVG)	https://developer.mozilla.org/	DOM manipulation, localStorage, FileReader, and 2D SVG rendering
Node.js Official Documentation	https://nodejs.org/en/docs	Server-side JavaScript runtime environment
Express.js Official Documentation	https://expressjs.com/	REST API routing, static file serving, and middleware
Multer (npm Documentation)	https://www.npmjs.com/package/multer	Multipart/form-data room image upload & MIME filtering
GeeksforGeeks — Software Engineering	https://www.geeksforgeeks.org/	SDLC models, DFD, UML diagrams, and manual testing
CodeWithHarry	https://www.youtube.com/@CodeWithHarry	Web development and Node.js backend tutorials
W3Schools — SVG & CSS Grid Tutorials	https://www.w3schools.com/	Responsive grid layouts and interactive SVG elements